



A+ LAB SERIES V5

Lab 18: Deploy a Virtual Machine with Hyper-V

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Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	4.1: Explain virtualization concepts.
CompTIA A+ (220-1202) Exam Objectives	1.2: Given a scenario, perform OS installations and upgrades in a diverse environment. 1.4: Given a scenario, use Microsoft Windows operating system features and tools.

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Introduction

In this lab, you will explore virtualization, where one or more virtual computers can operate inside one physical computer. Virtual machines that support separate guest operating systems are a popular way to run a real-time operating system alongside a favorite operating system, such as Linux or Windows. It allows users to experience different operating systems for training and testing, run software that their main operating system cannot, and work in a safe, sandboxed environment. There are many options for virtualization software, and it often needs to be downloaded and installed on the host before creating a virtual machine (VM). Additionally, you need an iso file for the OS you intend to install. An ISO file is a single file that contains the identical content of an optical disc, including the folder and file hierarchy. In this exercise, the contents of that file are the Ubuntu Linux distribution software.

Objective

In this lab, you will install a Debian Linux OS as a VM using Hyper-V, a virtualization platform. The virtualization software is available for installation, and the Debian ISO is also provided.

- Use PowerShell to verify Hyper-V is available.
- Enable Hyper-V using PowerShell.
- Verify that Hyper-V is enabled through Windows Features.
- Create a Linux virtual machine using Hyper-V and the Debian iso.

Lab Topology



Lab Settings

The information in the table below is needed to complete the lab. The task sections below provide details on how to use this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Creating a Virtual Machine Using Hyper-V

Hyper-V is a virtualization layer (hypervisor) installed on Windows 11 that creates virtual machines. It can run not only Windows virtual machines but also Linux distributions. Running Linux on a virtual machine in Windows can be beneficial for many reasons. Some of the more common reasons are:

- Testing a Linux distribution
- Testing software on multiple platforms
- Training purposes
- Reducing hardware costs and requirements

Virtualization provides many opportunities for an organization to explore and expand. Hyper-V is available on 64-bit versions of *Windows Professional*, *Enterprise*, and *Education* in *Windows 8* and later. It is not available on *Windows Home* Edition.

In this exercise, you will be installing Hyper-V and the Debian OS. Having the correct *Windows 11* version is not the only thing your PC needs; there are also hardware requirements:

- 64-bit CPU with Second Level Address Translation (SLAT).
- The processor must support VM Monitor Mode Extension (VT-c on Intel chips).
- 4GB of memory at a minimum.

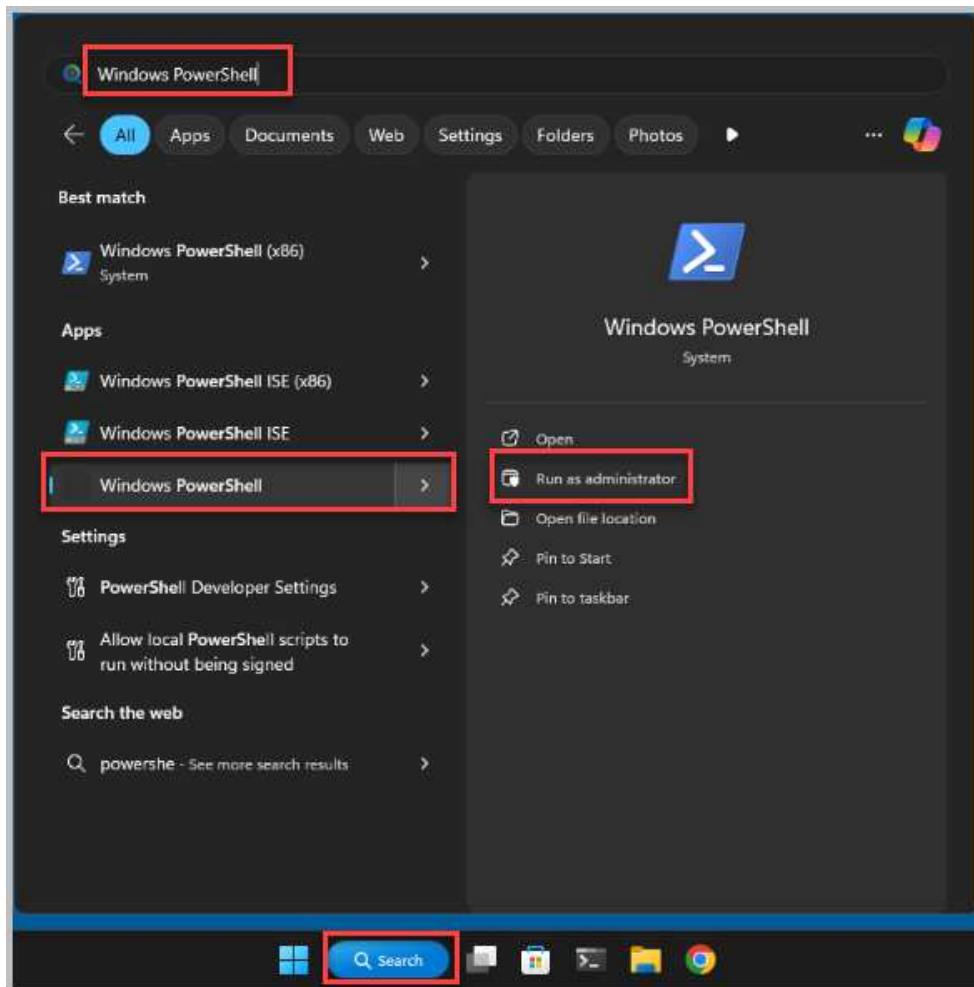
1.1 Verifying that Hyper-V is Available

1. Open a console connection to the **Win11-OS** system.



To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

- In the taskbar, click on the **Search** icon. Type **PowerShell**, select **Windows PowerShell** from the results, and select **Run as Administrator**.



- Click **Yes** on the *UAC* prompt.



- At the prompt, type the command below, followed by pressing the **Enter** key.

```
systeminfo.exe
```

```
PS C:\WINDOWS\system32> systeminfo.exe
```

- Scroll down to the bottom and locate the *Hyper-V Requirements* section to check if the PC can run Hyper-V. If this says “No hypervisor has been detected,” the system cannot run Hyper-V. You can see on this machine that a hypervisor has been detected, indicating that the lab machine can run it.

```

IP Address(es)
[01]: 192.168.1.11
[02]: f-88-rev6d-1u01-0726-320
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
PS C:\WINDOWS\system32>

```

- Check if the Hyper-V feature is enabled by entering the command below.

```
Get-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V -All
```

If you see output similar to the one below, it indicates that the Hyper-V feature is disabled.

```

PS C:\Users\systadmin> Get-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V-All

FeatureName      : Microsoft-Hyper-V-All
DisplayName       : Hyper-V
Description       : Provides services and management tools for creating and running virtual machines and their
                   resources.
RestartRequired  : Possible
State             : Disabled
CustomProperties  :

```

If you see output similar to the one below, it indicates that the Hyper-V feature is enabled.

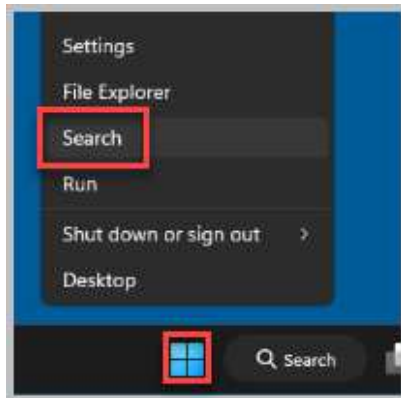
```

PS C:\Users\systadmin> Get-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V-All

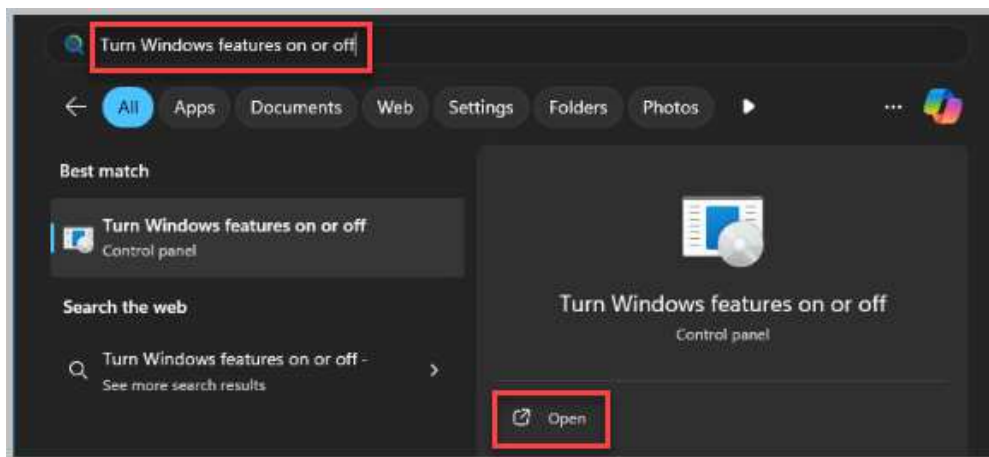
FeatureName      : Microsoft-Hyper-V-All
DisplayName       : Hyper-V
Description       : Provides services and management tools for creating and running virtual machines and their
                   resources.
RestartRequired  : Possible
State            : Enabled
CustomProperties  :

```

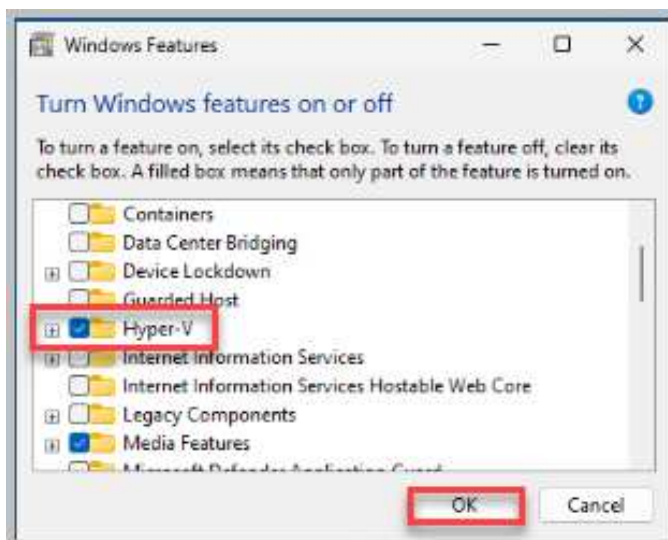

7. You can also check to see if the *Hyper-V* feature is enabled using the Graphical User Interface (GUI). Right-click the **Windows** icon on the taskbar and select **Search**.



8. Type **Turn Windows features on or off**. Select **Open**.



9. Verify that a check is in the checkbox next to *Hyper-V* in the *Windows Features* dialog box. Click **OK**.





If Hyper-V was not previously enabled, this box would not be checked, and you would need to check it and reboot the machine to apply the change.

10. Leave *Win11-OS* open and continue to the next task.

1.2 Enabling Hyper-V

Complete the following steps to enable *Hyper-V*.

1. In the *PowerShell* prompt, enter the command below. Notice that the system needs to be restarted to take effect. Press **Y**, then press **Enter** to reboot.

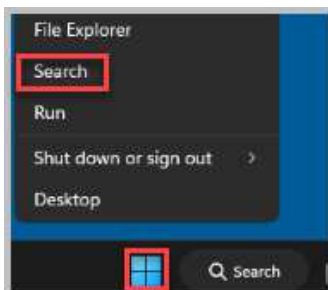
```
Enable-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V -All
```

```
PS C:\Users\systadmin> Enable-WindowsOptionalFeature -Online -FeatureName:Microsoft-Hyper-V-All
Do you want to restart the computer to complete this operation now?
[Y] Yes [N] No [?] Help (default is "Y"): Y
```

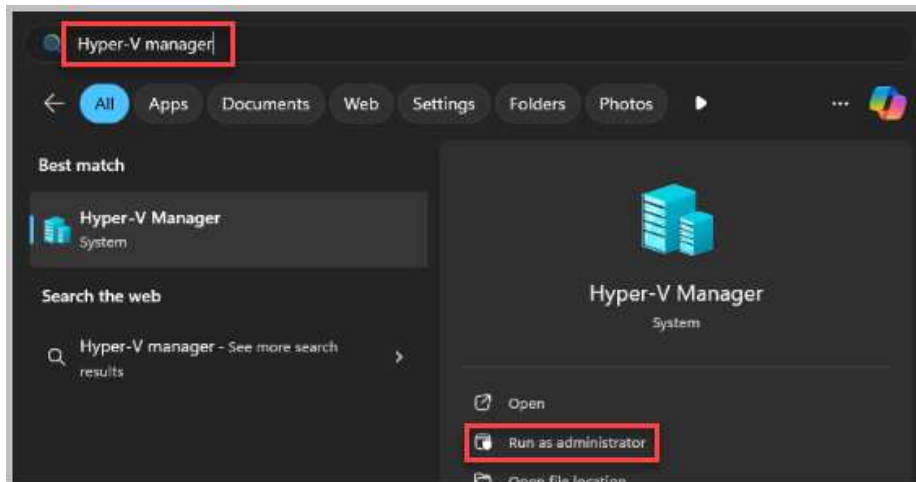
2. After the reboot, leave *Win11-OS* open for the next lab activity.

1.3 Creating a Virtual Machine

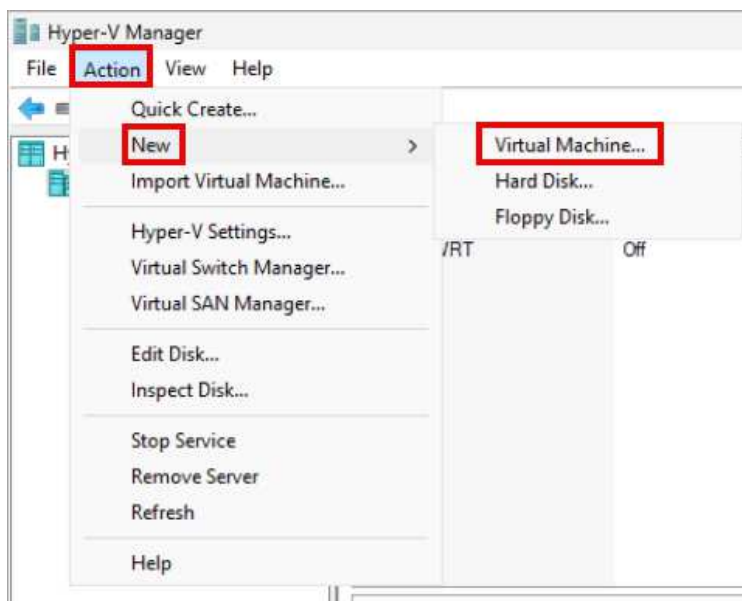
1. Right-click on the **Start** button and select **Search**.



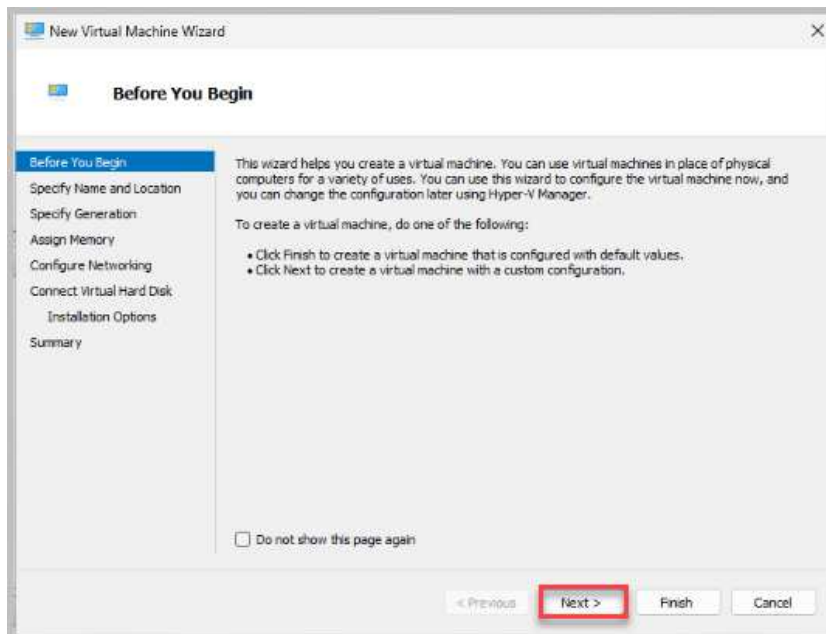
2. Type **Hyper-V manager** and select **Run as Administrator**.



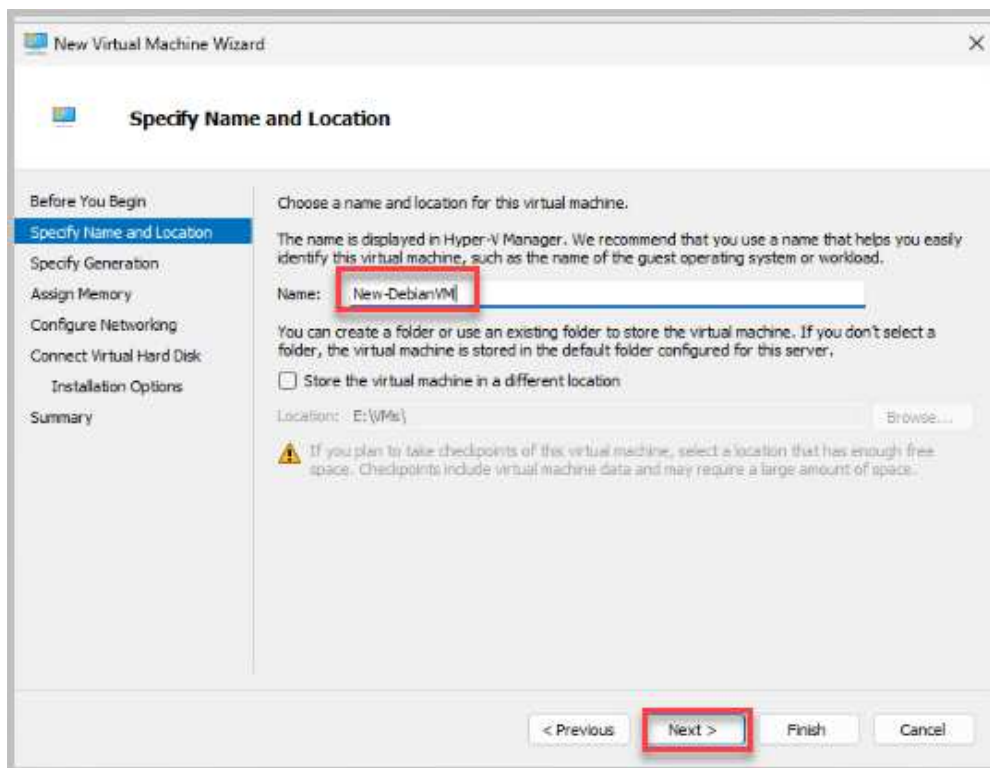
3. In the *Hyper-V Manager* window, click on the **Action** menu. Select **New** and click on **Virtual Machine**.



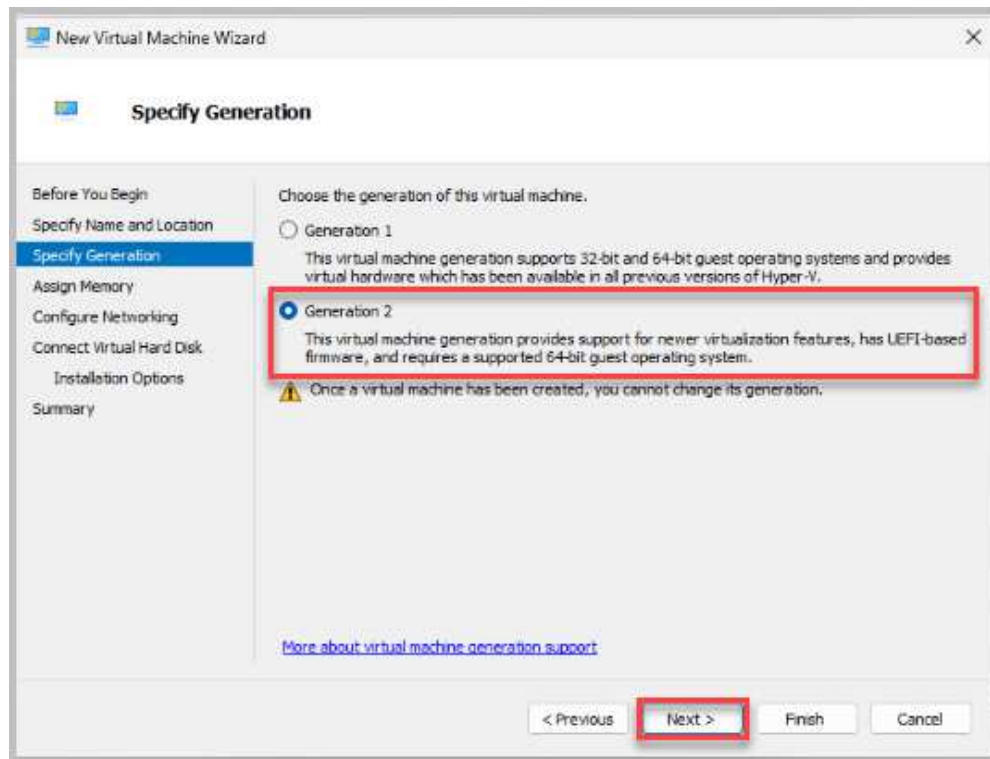
4. In the *New Virtual Machine Wizard* window, click the **Next** button.



5. On the *Specify Name and Locations* step, type **New-DebianVM** in the *Name* text field. Click **Next**.

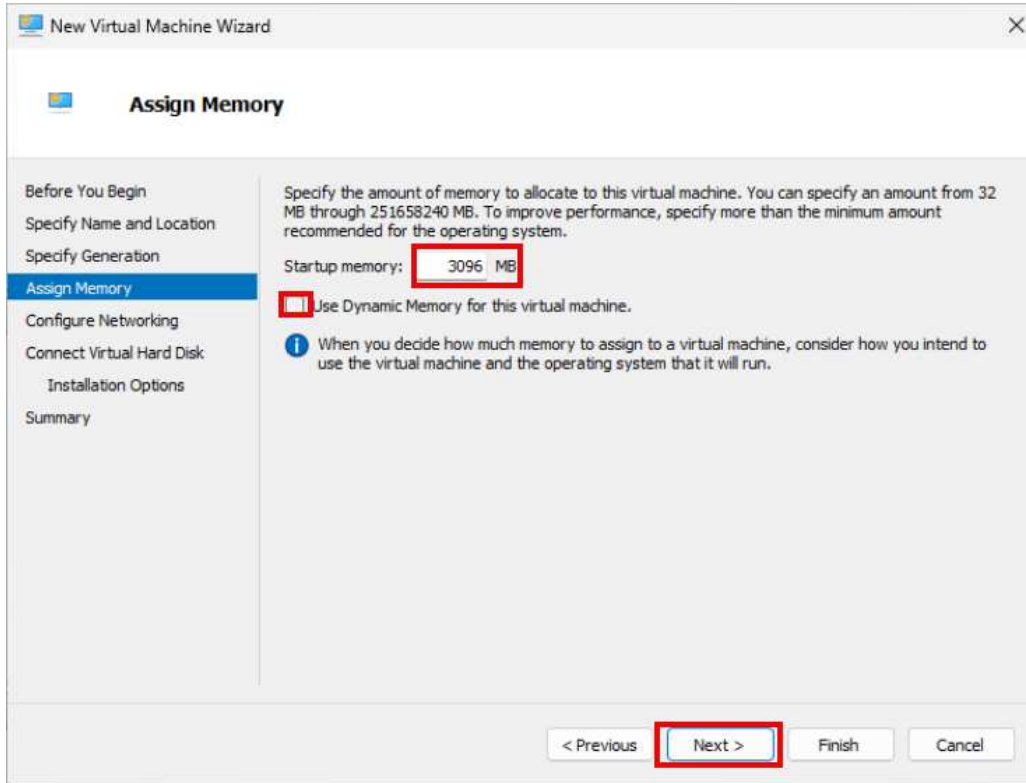


6. On the *Specify Generation* step, select the **Generation 2** option. Click **Next**.



Generation 1 and 2 VMs differ in their capabilities and compatibility. Gen 1 is better suited for older operating systems and legacy hardware, while Gen 2 supports modern hardware features, offers better performance, and is ideal for newer operating systems. For this installation, we'll use a Generation 2 VM. When using "Generation 2" in Hyper-V, it automatically configures the VM to use "Secure Boot". Since the host does not have Secure Boot enabled, you will need to disable it. You'll simply right-click on the VM > Settings > Security > Disable Secure Boot. After that, you can connect and power on the VM and go through the installation as normal.

- On the *Assign Memory* step, select the amount of RAM to allocate for your virtual machine. Type **3096** in the Startup memory field to configure 3096 MB of RAM for the new VM. **Uncheck** the checkbox next to **Dynamic Memory**. Click **Next**.



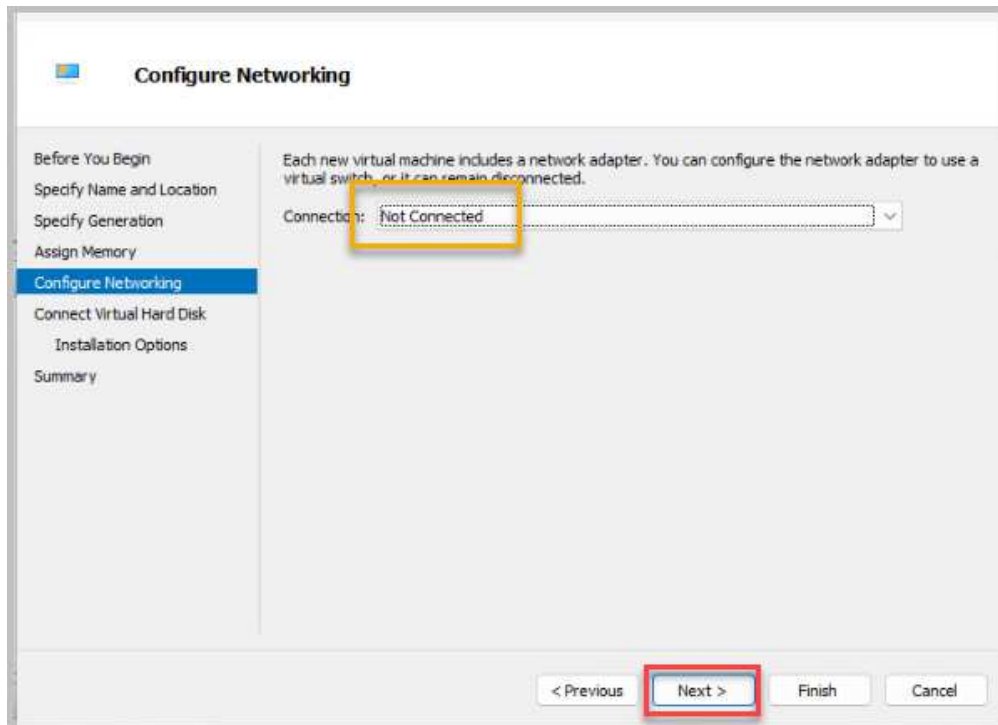
The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Assign Memory' step. On the left, a list of steps includes 'Before You Begin', 'Specify Name and Location', 'Specify Generation', 'Assign Memory' (which is selected and highlighted in blue), 'Configure Networking', 'Connect Virtual Hard Disk', 'Installation Options', and 'Summary'. The main area contains instructions: 'Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 32 MB through 251658240 MB. To improve performance, specify more than the minimum amount recommended for the operating system.' Below this, the 'Startup memory' is set to '3096 MB'. A checkbox labeled 'Use Dynamic Memory for this virtual machine.' is present and is unchecked. An information icon with a blue 'i' is followed by the text: 'When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.' At the bottom, there are four buttons: '< Previous', 'Next >' (which is highlighted with a red box), 'Finish', and 'Cancel'.



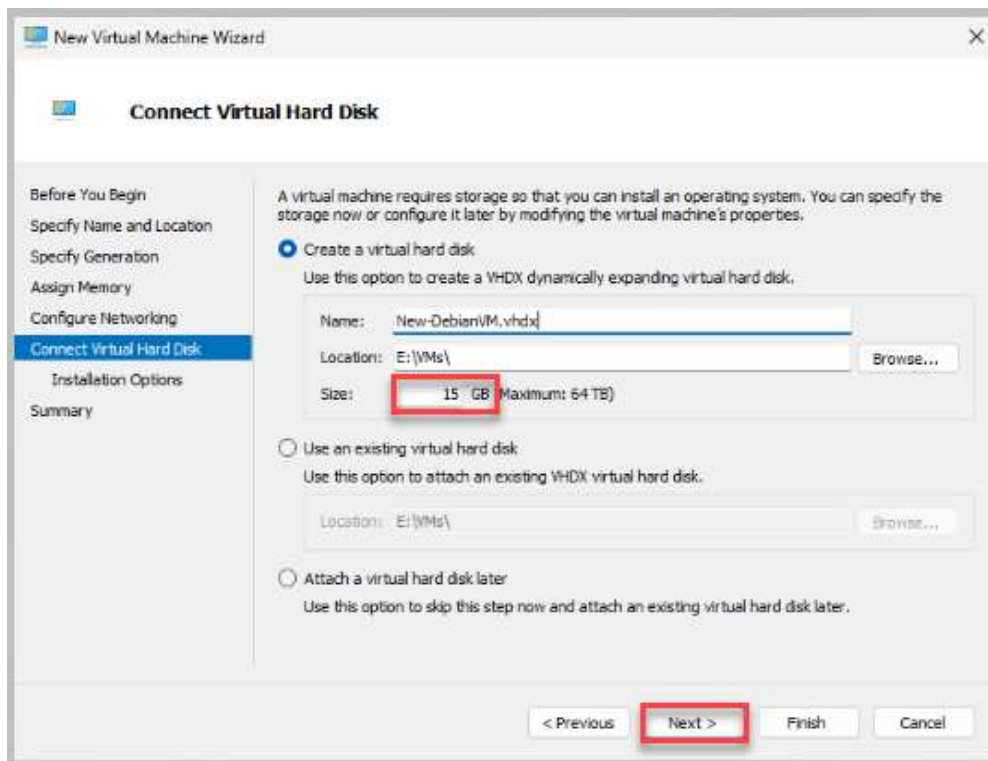
Using dynamic memory for your virtual machine allows it to automatically adjust its memory usage based on its needs. This means it can borrow unused memory from other VMs when it needs more, and return it when it doesn't, optimizing resource utilization.

By not using dynamic memory, you allocate a fixed amount of RAM to your VM, leading to predictable performance. This can simplify management and potentially improve performance in resource-constrained environments. However, it can also result in resource waste if the VM doesn't consistently use all the allocated memory, and limits flexibility in adapting to changing workloads.

8. On the *Configure Networking* step, leave the default selection of **Not Connected** as this virtual machine will not be connected to any other virtual machines or physical machines. Select **Next**.



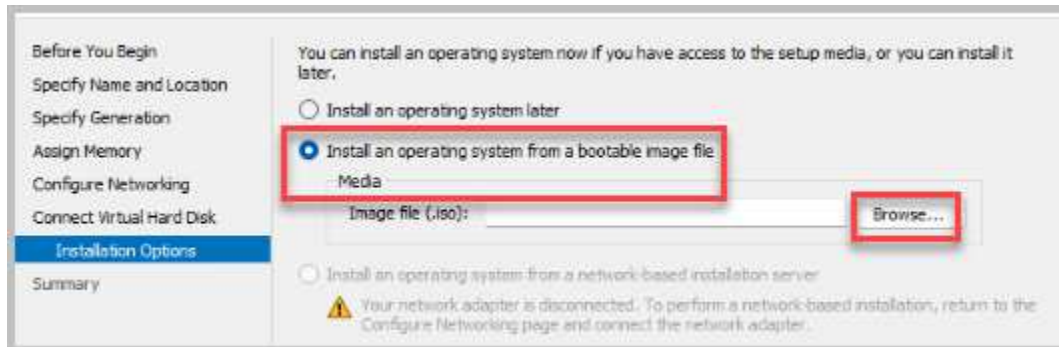
9. On the *Connect Virtual Hard Disk* step, change the Size setting to **15 GB**. Click **Next**.



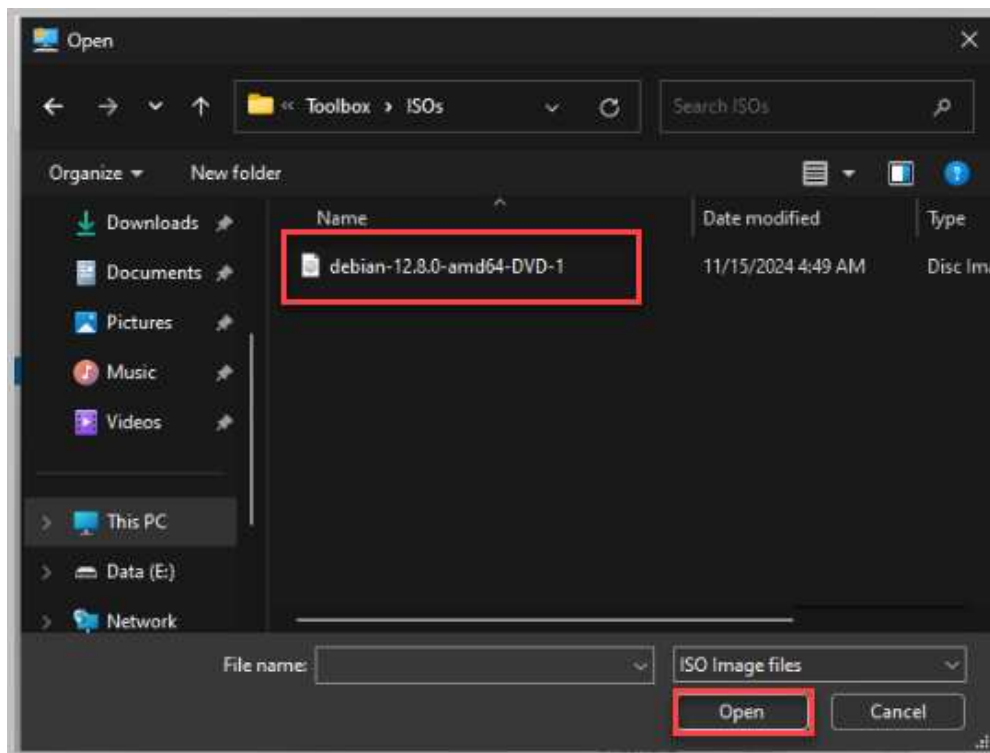


A minimal Ubuntu installation can be achieved with a modest 8.6 GB of disk space. However, for a comprehensive setup including recommended software and future growth, Ubuntu officially recommends 25 GB. For a seamless user experience with ample room for additional files and applications, we suggest aiming for at least 50 GB of free disk space.

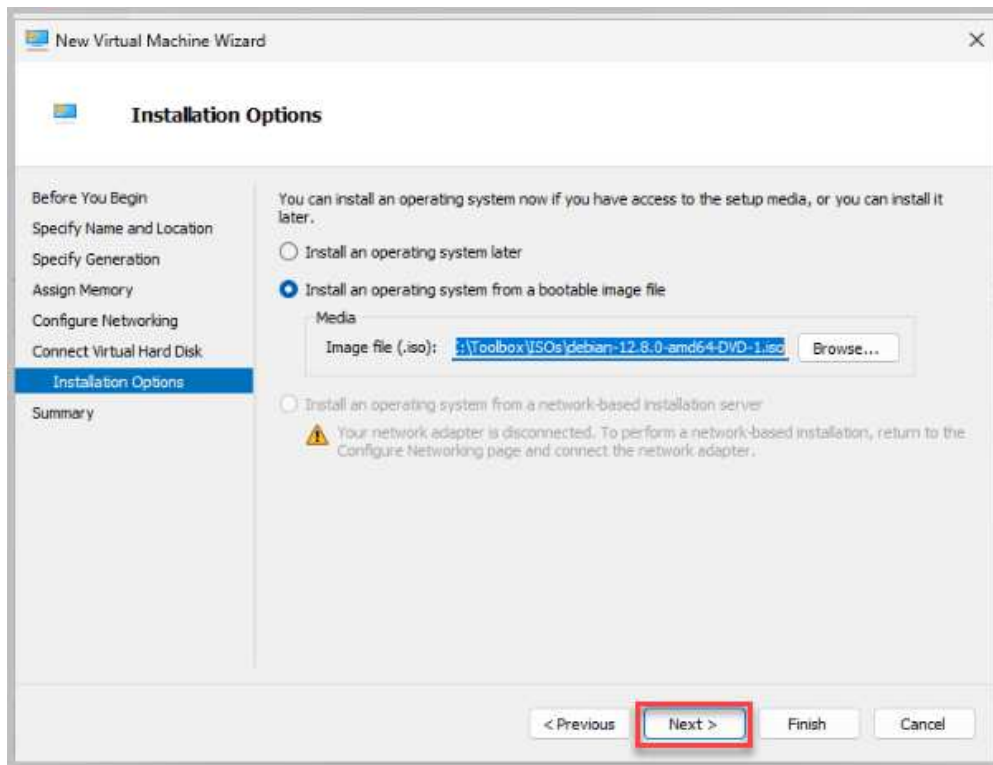
10. On the *Installation Options* step, select the **Install an operating system from a bootable image file** option. Select the **Image file (.iso)** radio button and select **Browse**.



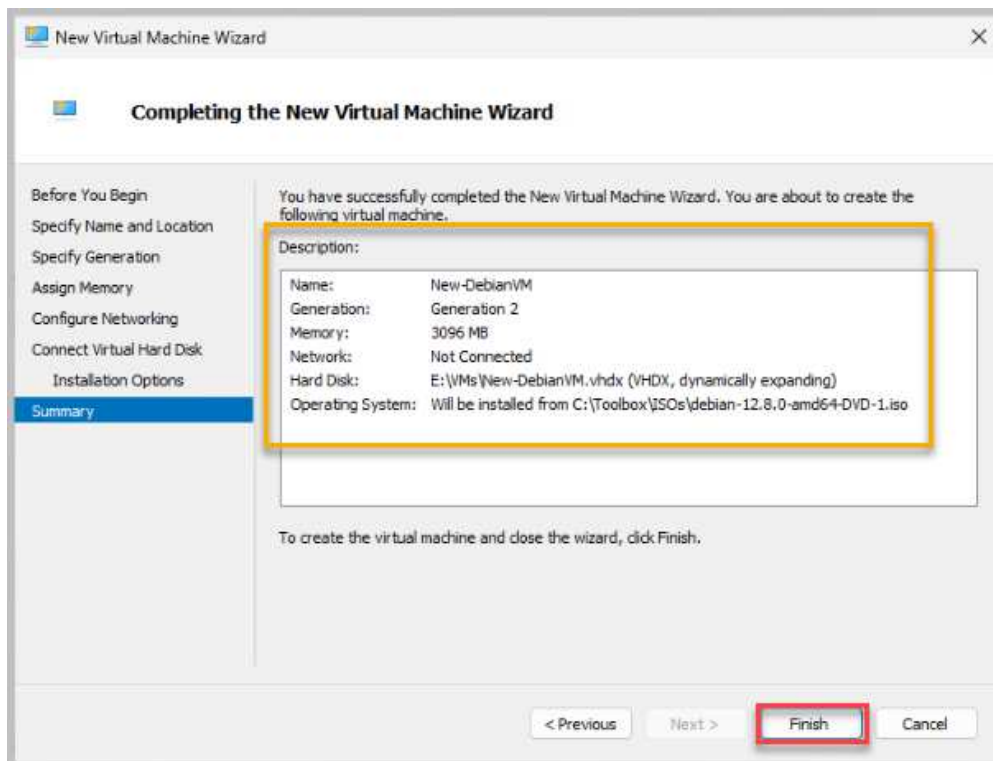
11. Navigate to the path for the ISO file with the *Lubuntu* installation files (C:\Users\sysadmin\Desktop\ToolBox\ISOs\debian-12.8.0-amd64-DVD-1.iso). Select the **Debian-12.8.0-amd64-DVD-1.iso** file and click **Open**.



12. Back on the *New Virtual Machine Wizard*, confirm your selections, and click **Next**.



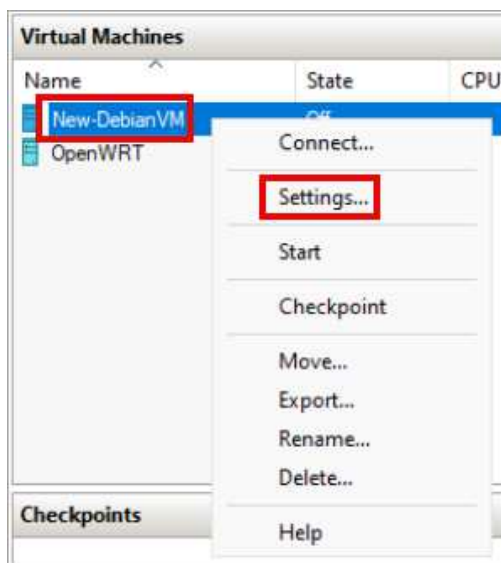
13. On the *Completing the New Virtual Machine Wizard* step, review the settings and click **Finish**.



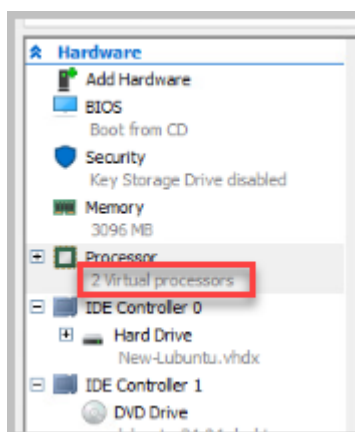
14. You will now see your newly deployed virtual machine in the list of virtual machines within the *Hyper-V Manager*.



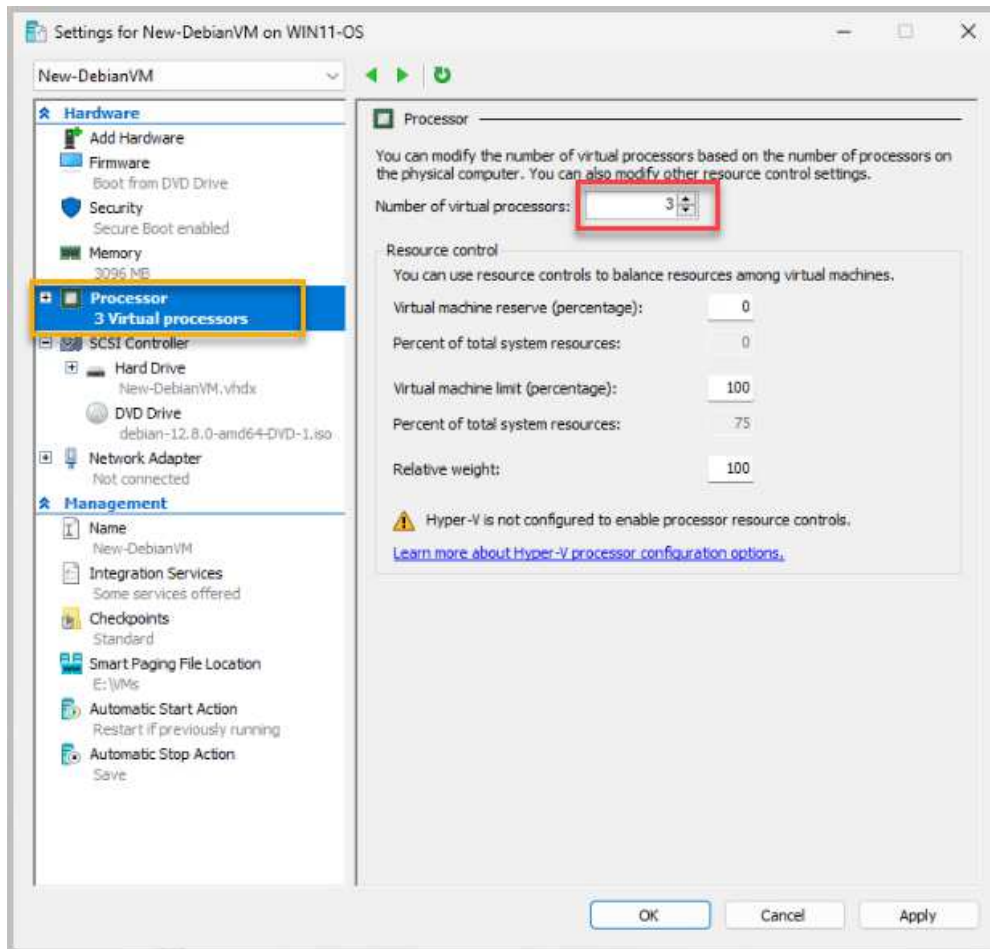
15. Right-click the *New-DebianVM* virtual machine and select **Settings**.



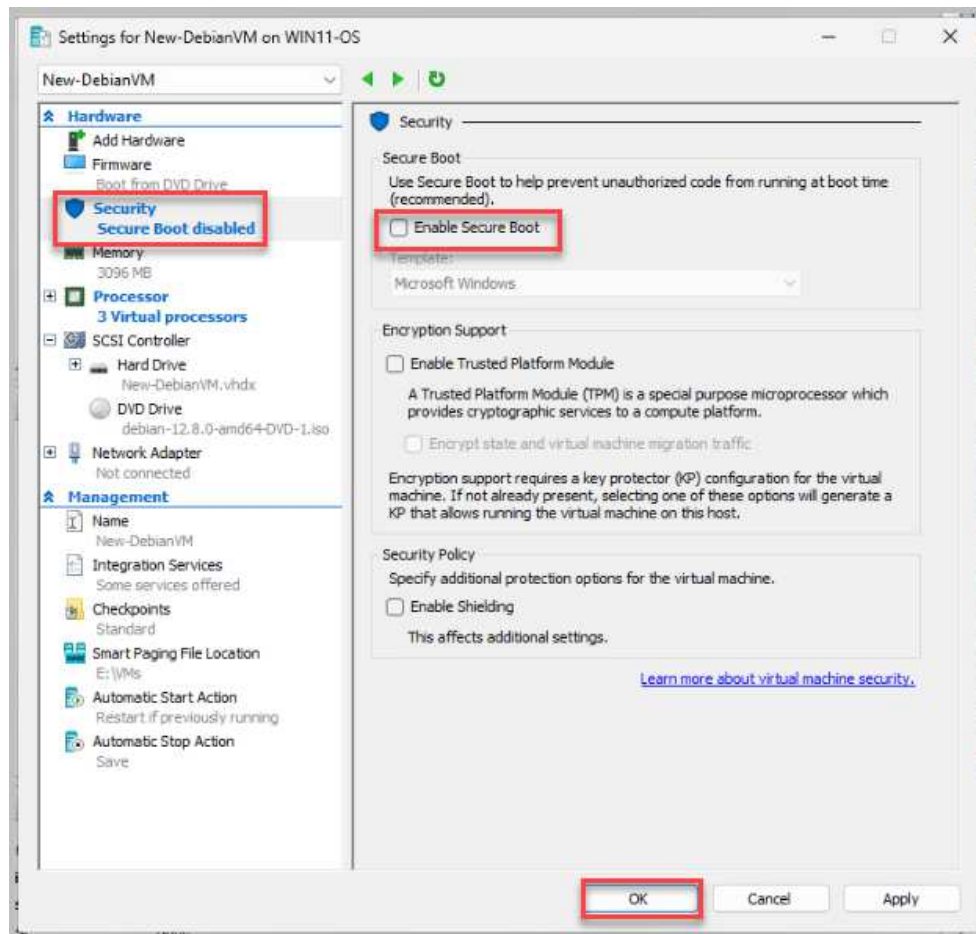
16. Under *Processor*, select **2 Virtual Processors**.



17. Change the *Number of Virtual Processors* to **3**. Notice the Processor totals changed to *3 Virtual Processors*.



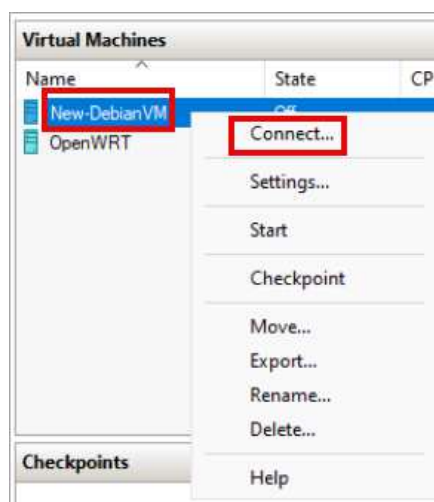
18. Select **Security-Secure Boot enabled** and **Uncheck** *Enable Secure Boot*. Select **OK**.



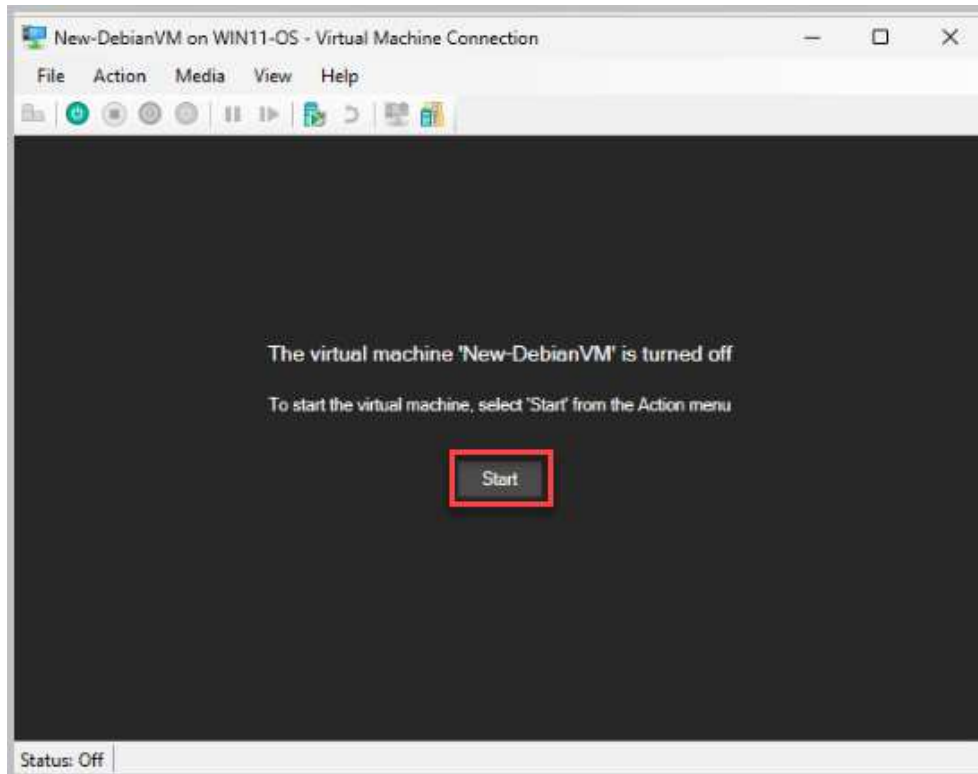
19. Leave the *Win11-OS* system open to continue with the next task.

1.4 Complete Installation of the Debian Virtual Machine

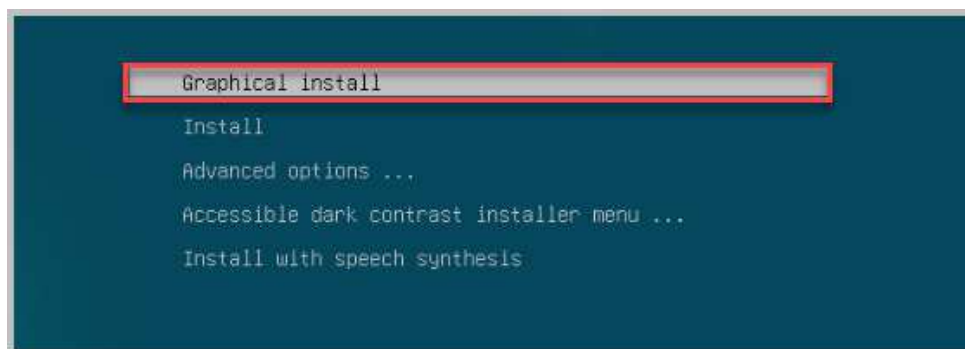
1. In the *Hyper-V Manager* window and underneath the *Virtual Machines* section, right-click on **New-DebianVM**, and select **Connect**.



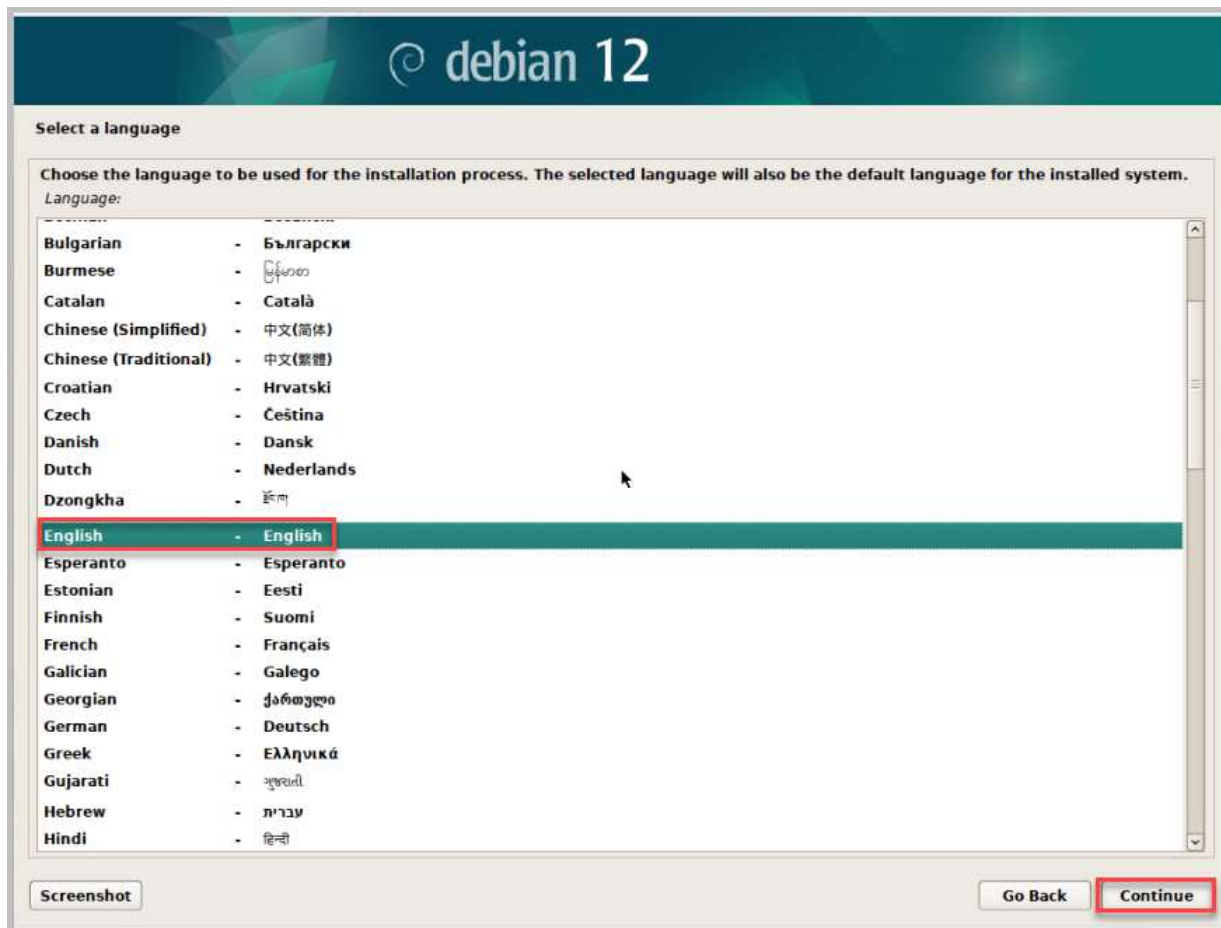
2. *Hyper-V* will now prompt you to start your virtual machine. Click the **Start** button to begin installing the operating system.



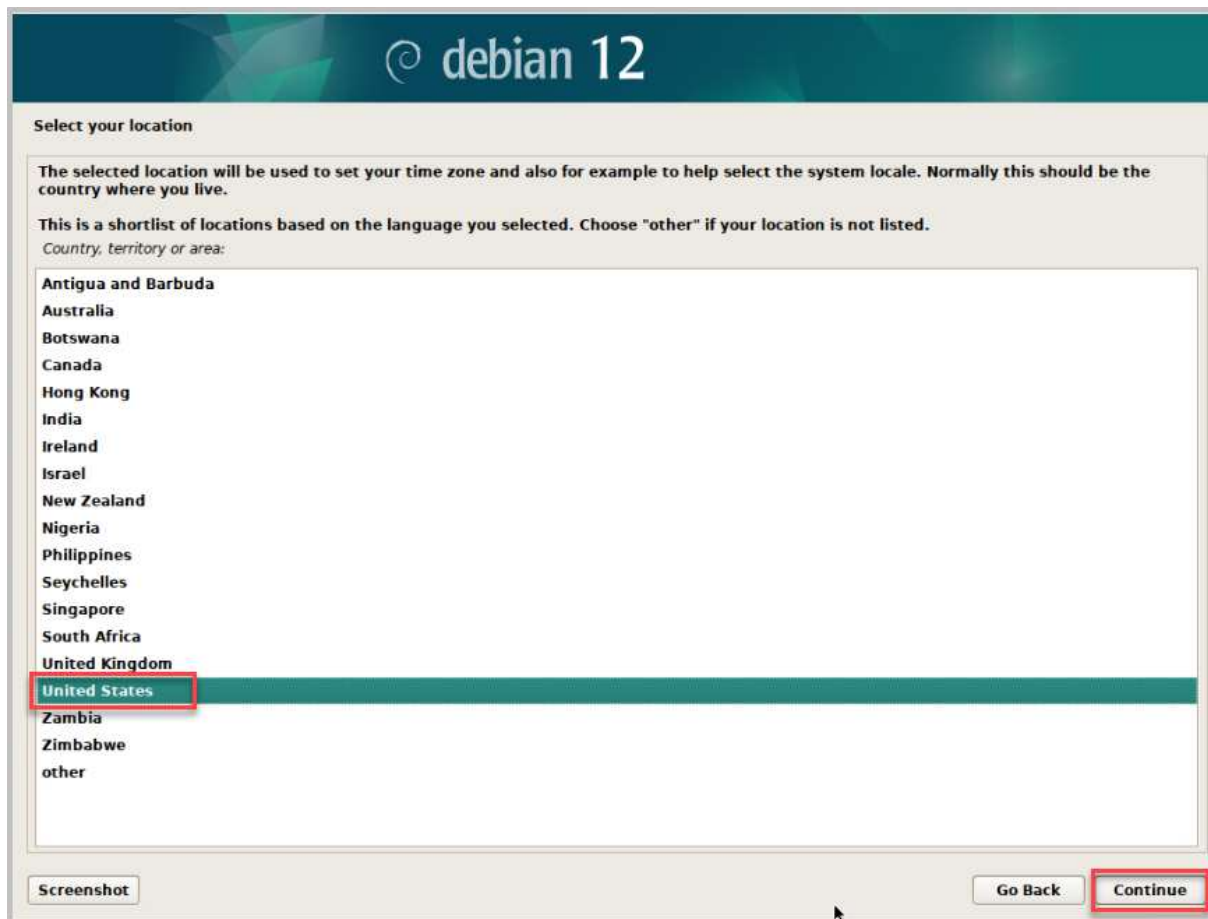
3. Select **Enter** on your keyboard to select *Graphical install*.



4. Leave the default selection of language, **English**. Select **Continue**.



5. Leave the default selection of location, **United States**. Select **Continue**.



The image shows the Debian 12 installation window for selecting a location. The window has a teal header with the Debian logo and "debian 12". Below the header, the text "Select your location" is displayed. A paragraph explains that the selected location will be used to set the time zone and help select the system locale. Another paragraph states that this is a shortlist of locations based on the language selected, and to choose "other" if the location is not listed. A list of countries and territories is shown, with "United States" highlighted in a teal bar and a red box. At the bottom, there are three buttons: "Screenshot", "Go Back", and "Continue". The "Continue" button is highlighted with a red box.

© debian 12

Select your location

The selected location will be used to set your time zone and also for example to help select the system locale. Normally this should be the country where you live.

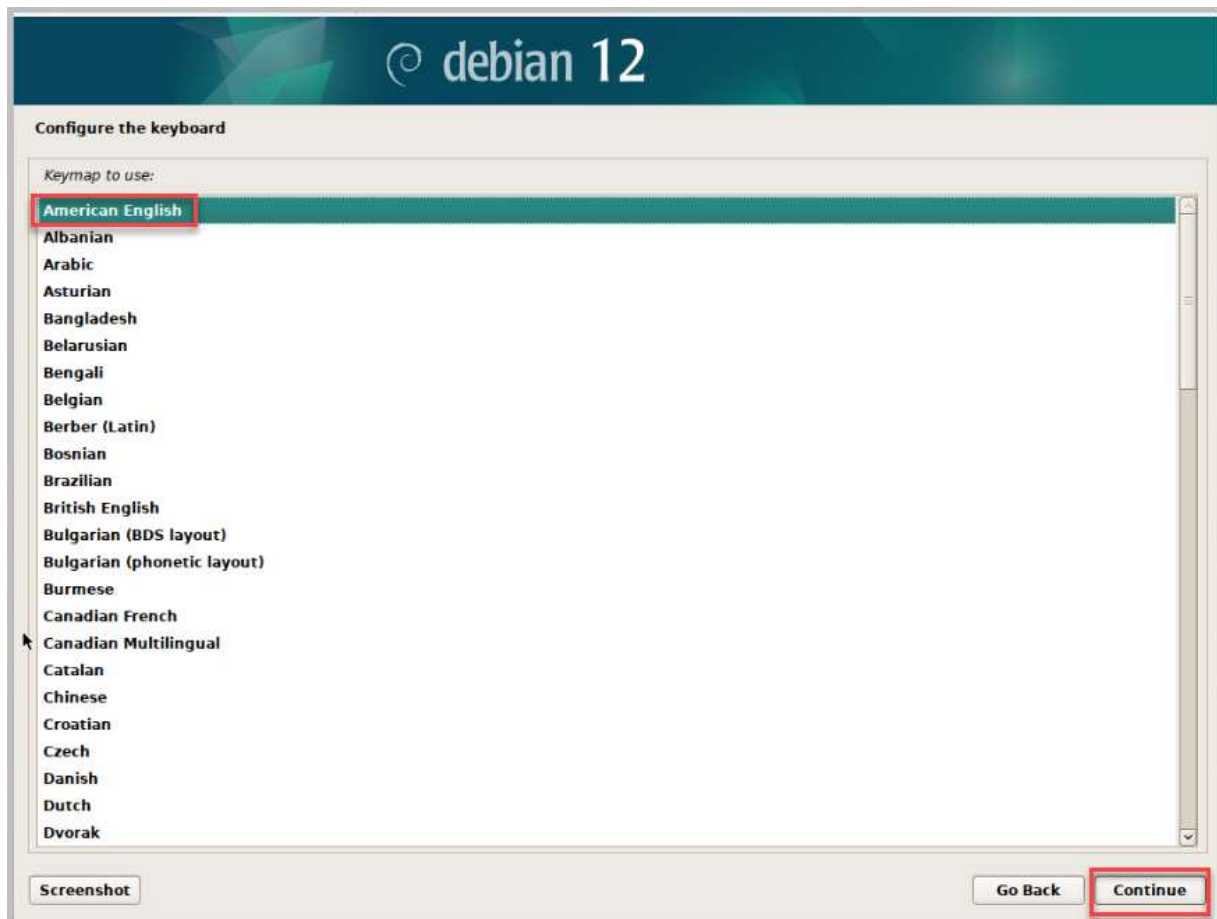
This is a shortlist of locations based on the language you selected. Choose "other" if your location is not listed.

Country, territory or area:

- Antigua and Barbuda
- Australia
- Botswana
- Canada
- Hong Kong
- India
- Ireland
- Israel
- New Zealand
- Nigeria
- Philippines
- Seychelles
- Singapore
- South Africa
- United Kingdom
- United States**
- Zambia
- Zimbabwe
- other

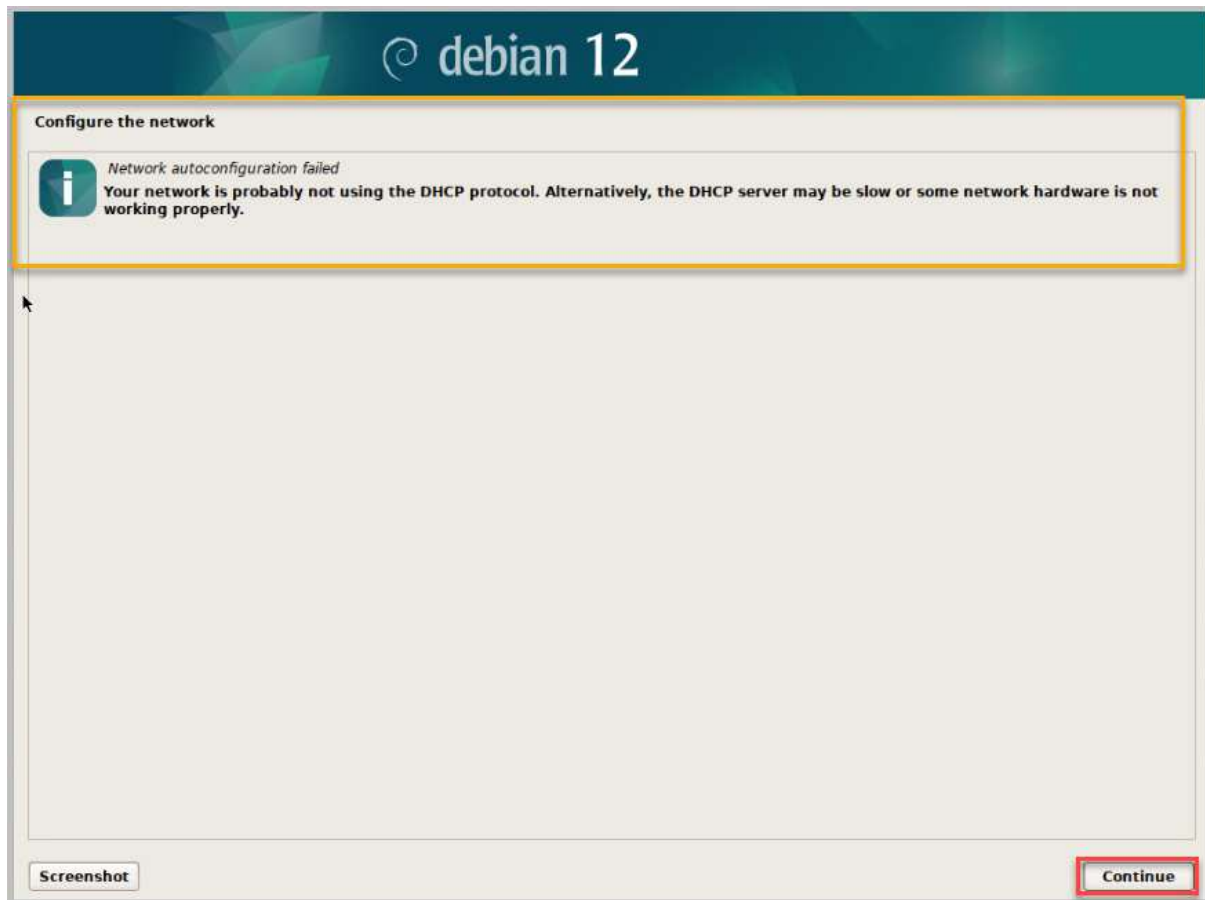
Screenshot Go Back Continue

6. Leave the default selection of keyboard, **American English**. Select **Continue**.

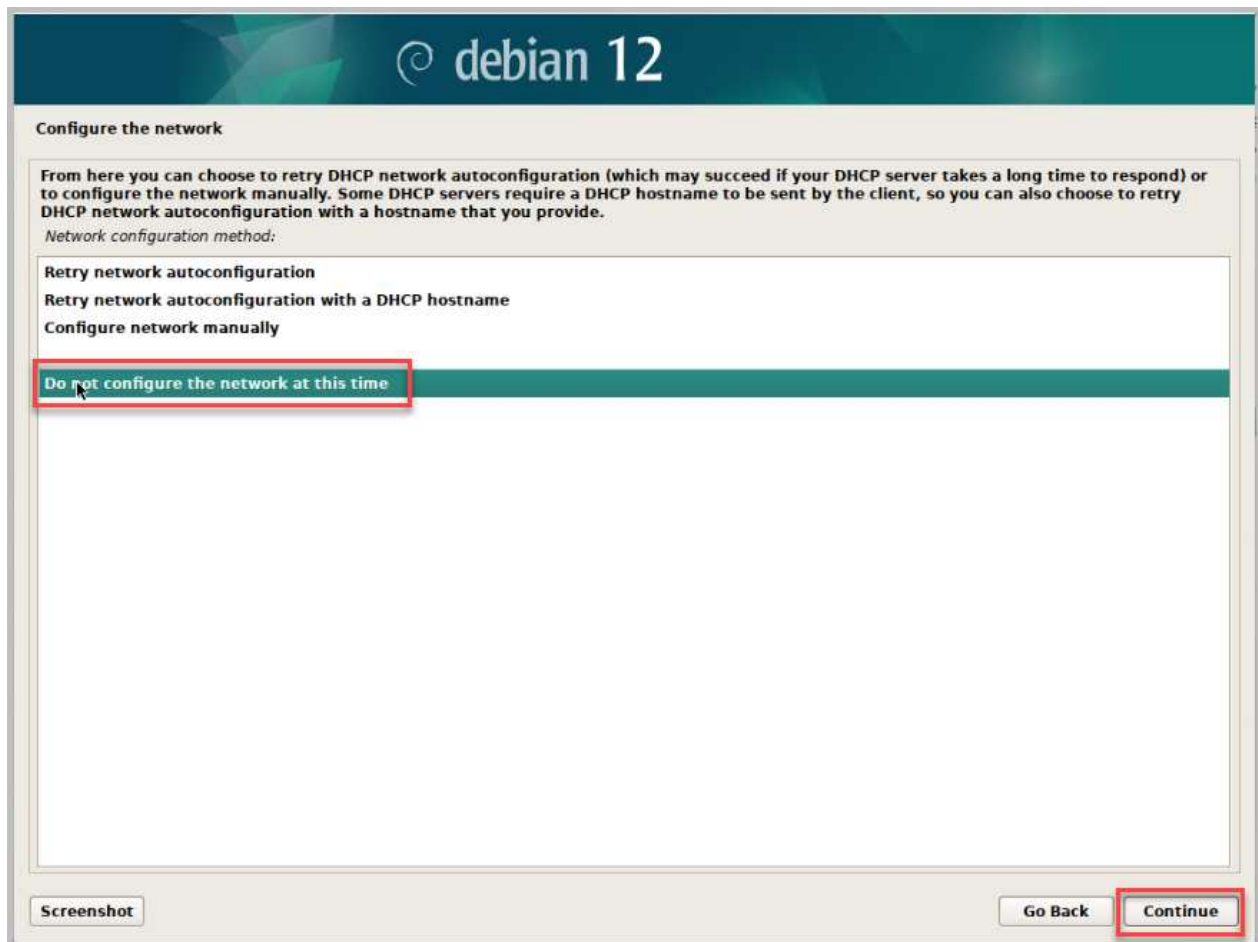


7. The setup will take a few minutes to complete. Please be patient.

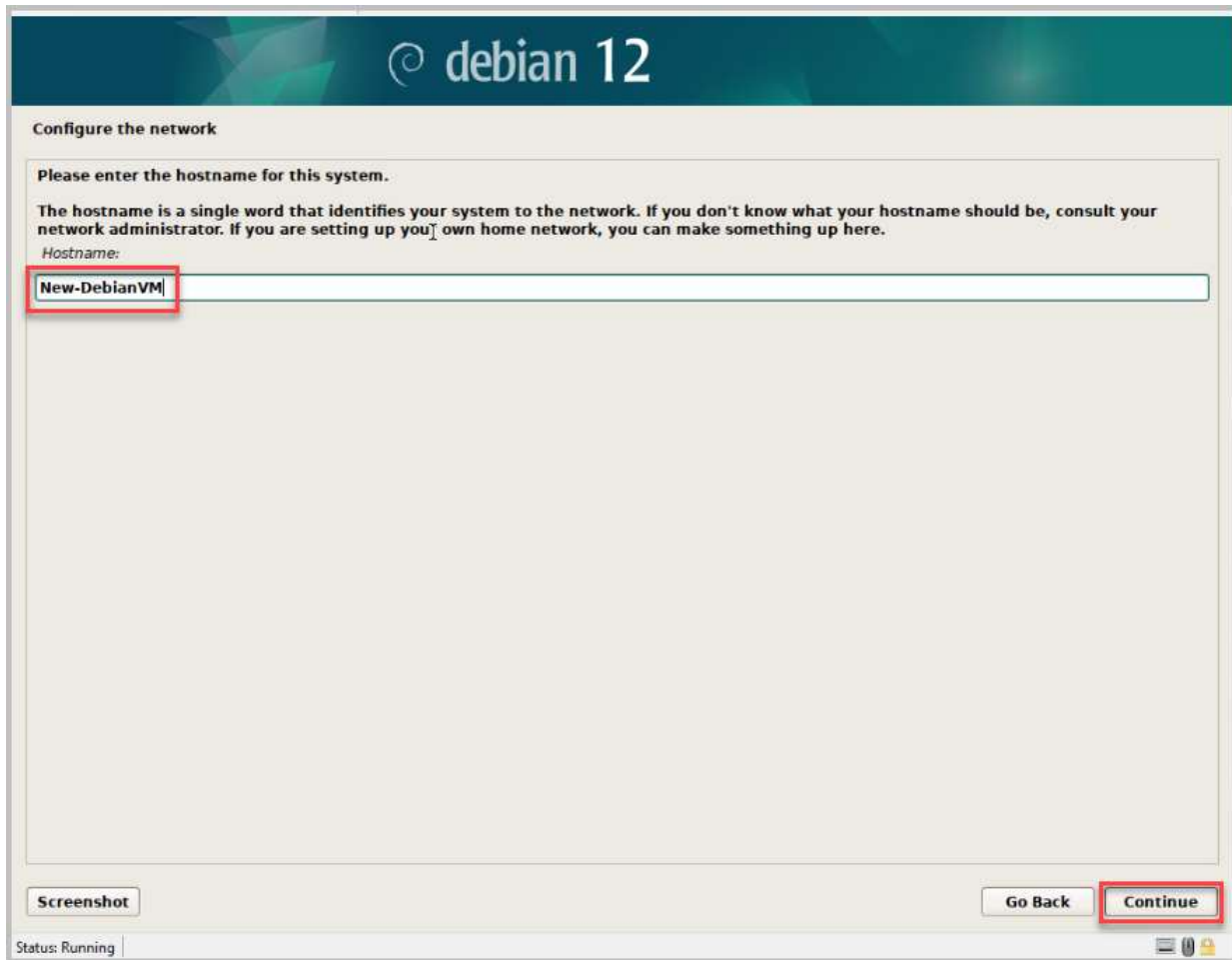
8. Ignore the Error *Network Autoconfiguration Failed* as you didn't select a network adapter when creating the New-DebianVM. Select **Continue**.



9. Select **Do not configure the network at this time**, and select **Continue**.



10. Type **New-DebianVM** for the *Hostname*. Select **Continue**.



The image shows the 'Configure the network' window in the Debian 12 installer. The window has a teal header with the Debian logo and 'debian 12'. Below the header, the title 'Configure the network' is displayed. The main content area contains the text: 'Please enter the hostname for this system.' followed by a detailed explanation of hostnames. Below this, the label 'Hostname:' is followed by a text input field containing 'New-DebianVM'. At the bottom of the window, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red box. In the bottom left corner, the status 'Status: Running' is shown. In the bottom right corner, there are icons for network, sound, and power.

debian 12

Configure the network

Please enter the hostname for this system.

The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.

Hostname:

New-DebianVM

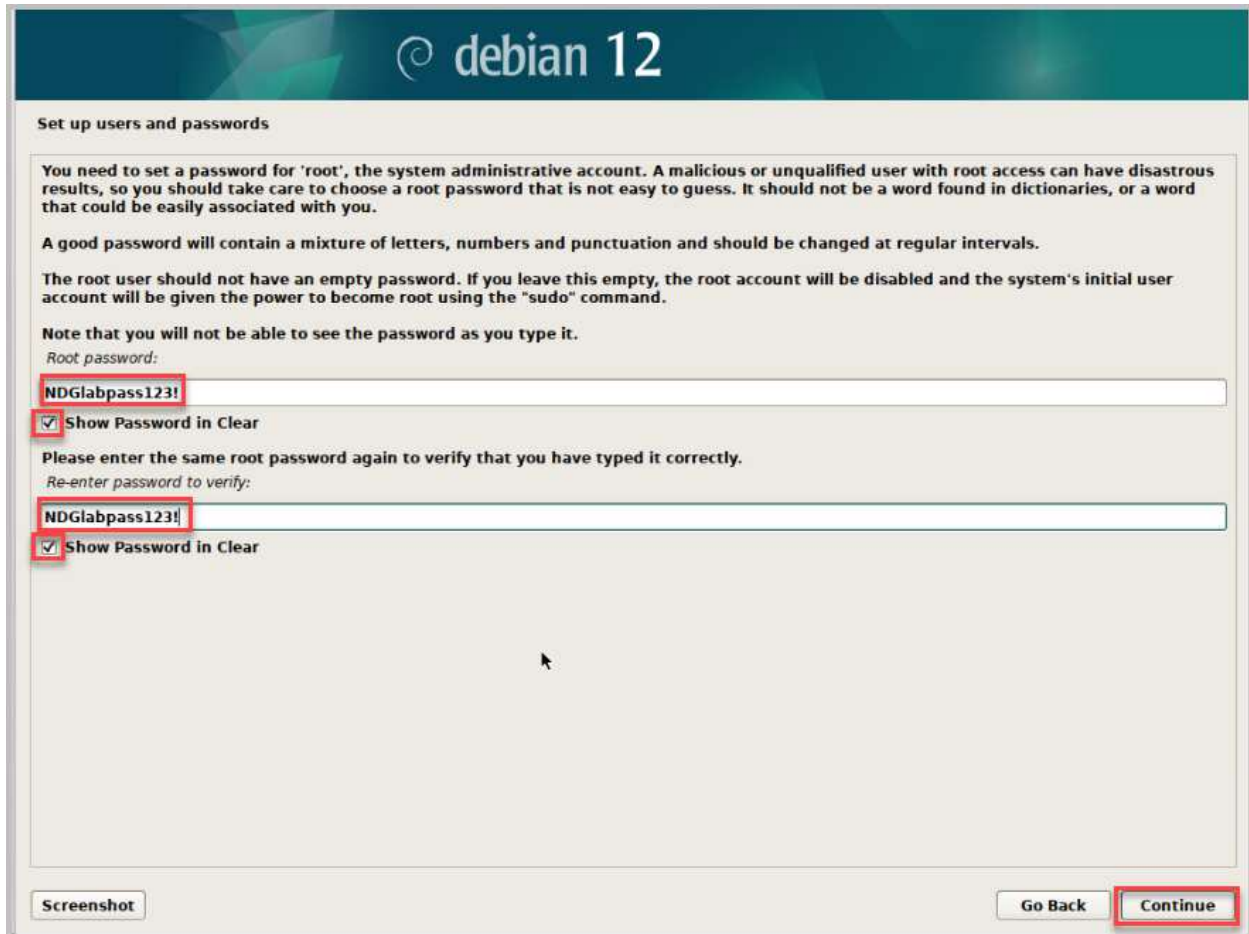
Screenshot

Go Back

Continue

Status: Running

11. Enter **NDGlabpass123!** for both the initial password and verification password. Select **Show Password in Clear**, which will show what you typed for the password. Select **Continue**.



The image shows the 'Set up users and passwords' screen in the Debian 12 installer. The window has a teal header with the Debian logo and 'debian 12'. The main content area is light gray and contains instructions for setting the root password. It includes a text input field for the root password, a checkbox for 'Show Password in Clear' which is checked, and a second text input field for re-entering the password. The 'Continue' button at the bottom right is highlighted with a red box. Other elements include a 'Screenshot' button at the bottom left and a 'Go Back' button at the bottom right.

Set up users and passwords

You need to set a password for 'root', the system administrative account. A malicious or unqualified user with root access can have disastrous results, so you should take care to choose a root password that is not easy to guess. It should not be a word found in dictionaries, or a word that could be easily associated with you.

A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.

The root user should not have an empty password. If you leave this empty, the root account will be disabled and the system's initial user account will be given the power to become root using the "sudo" command.

Note that you will not be able to see the password as you type it.

Root password:

NDGlabpass123!

☒ Show Password in Clear

Please enter the same root password again to verify that you have typed it correctly.

Re-enter password to verify:

NDGlabpass123!

☒ Show Password in Clear

Screenshot

Go Back Continue

12. Type **Student** in the textbox as the *Full name for the new user*. Select **Continue**.



The image shows the Debian 12 installer window for setting up users and passwords. The title bar says "debian 12". The main heading is "Set up users and passwords". Below this, a message states: "A user account will be created for you to use instead of the root account for non-administrative activities." Another message follows: "Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice." Below this, it says "Full name for the new user:". A text input field contains the word "Student". At the bottom left is a "Screenshot" button. At the bottom right are "Go Back" and "Continue" buttons. Red boxes highlight the "Student" text and the "Continue" button.

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Set up users and passwords

A user account will be created for you to use instead of the root account for non-administrative activities.

Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.

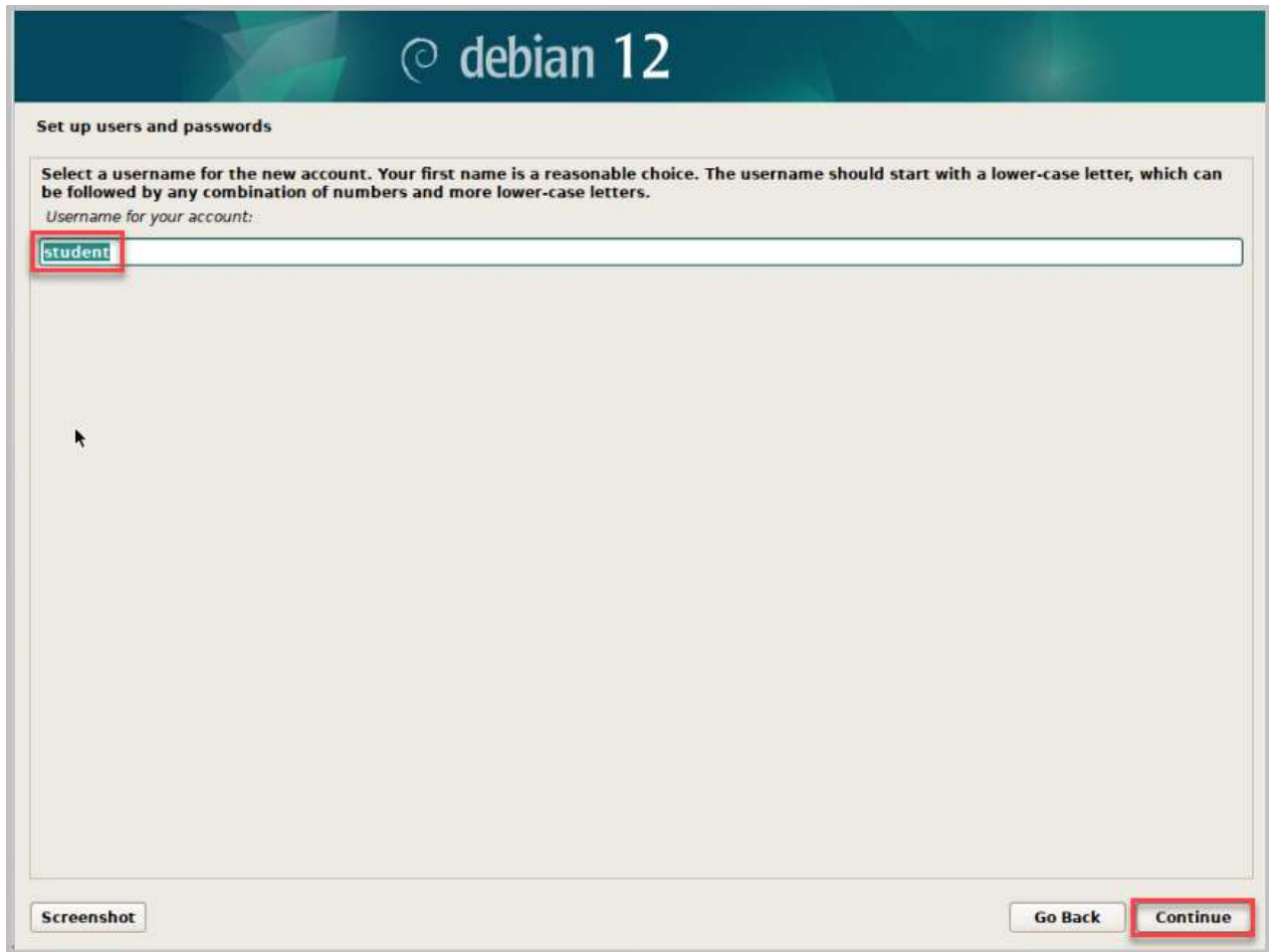
Full name for the new user:

Student

Screenshot

Go Back Continue

13. Leave the default username of **student**. Select **Continue**.



The image shows the 'Set up users and passwords' screen of the Debian 12 installer. The window has a teal header with the Debian logo and 'debian 12'. Below the header, the title 'Set up users and passwords' is displayed. The main text instructs the user to 'Select a username for the new account' and provides guidelines: 'Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.' Below this, it says 'Username for your account:' followed by a text input field containing the word 'student'. The 'student' text is highlighted with a red box. At the bottom of the window, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red box.

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Set up users and passwords

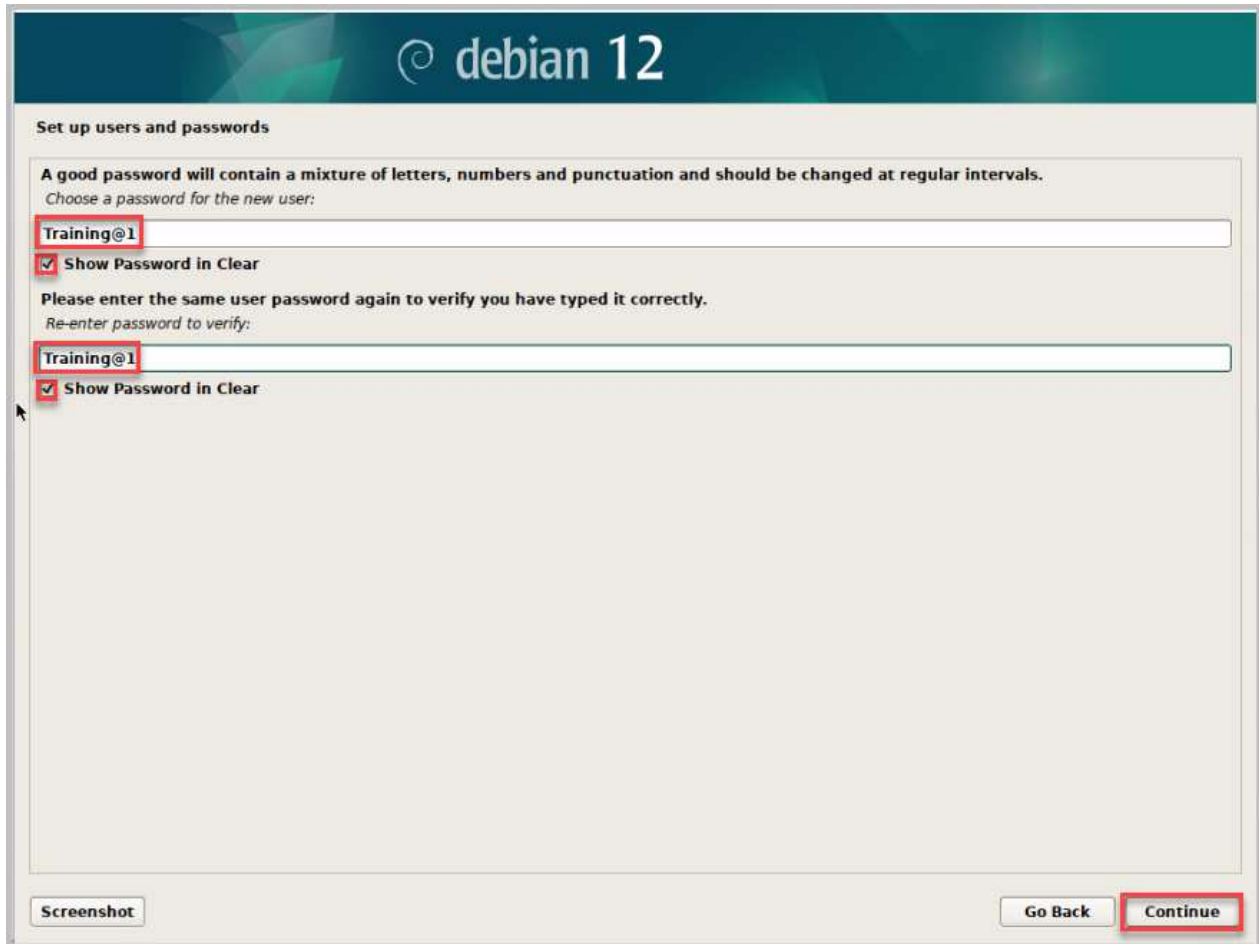
Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.

Username for your account:

student

Screenshot Go Back Continue

14. Enter **Training@1** for both the initial password and verification password. Select **Show Password in Clear**, which will show what you typed for the password. Select **Continue**.



The image shows the 'Set up users and passwords' screen in the Debian 12 installer. The header features the Debian logo and 'debian 12'. The main section is titled 'Set up users and passwords'. It includes a note about password strength: 'A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals. Choose a password for the new user:'. Below this, the password 'Training@1' is entered in a text field, and the checkbox 'Show Password in Clear' is checked. A second instruction says 'Please enter the same user password again to verify you have typed it correctly. Re-enter password to verify:'. The same password 'Training@1' is entered in the second text field, and the 'Show Password in Clear' checkbox is also checked. At the bottom, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a red box.

© debian 12

Set up users and passwords

A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.
Choose a password for the new user:

Training@1

☒ Show Password in Clear

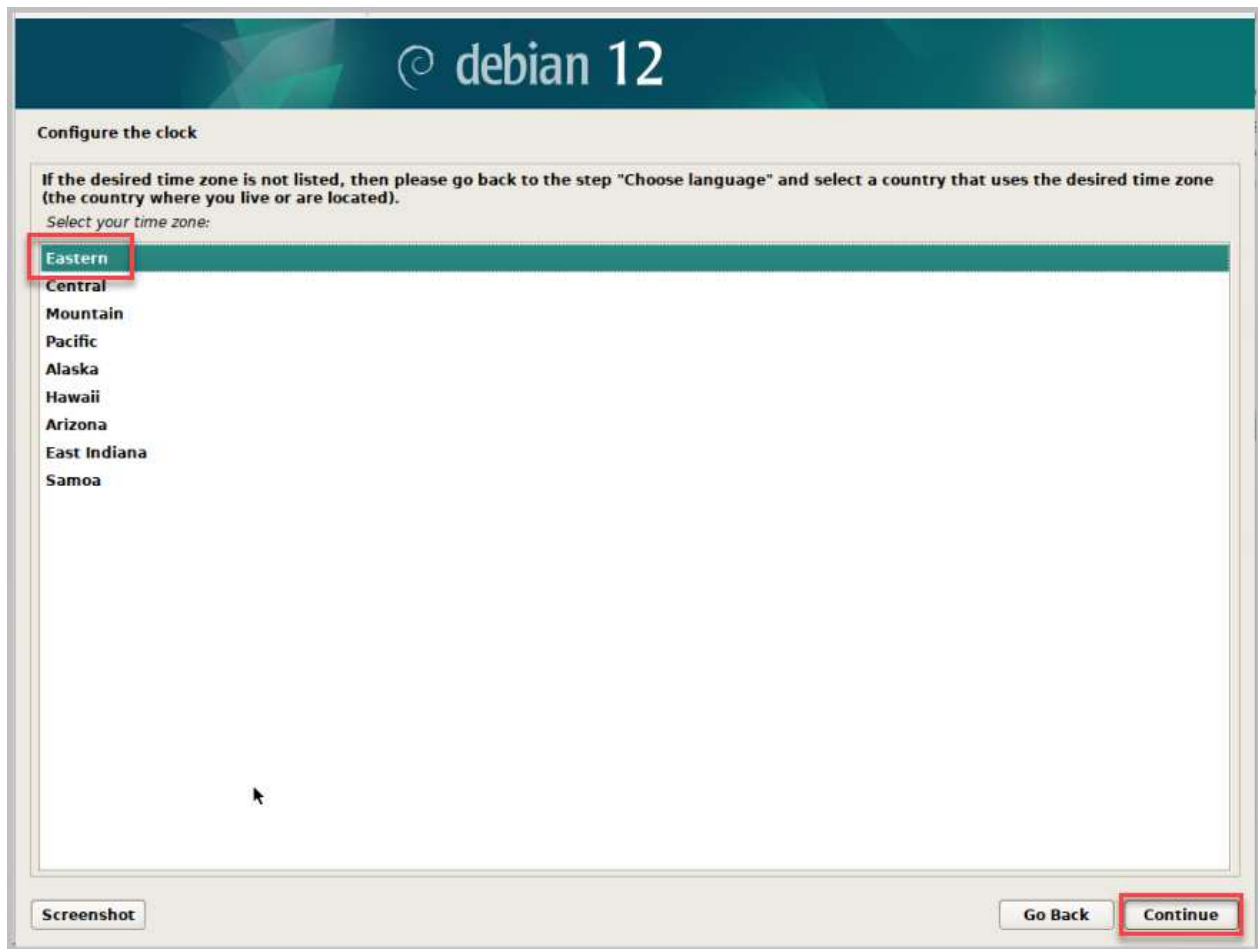
Please enter the same user password again to verify you have typed it correctly.
Re-enter password to verify:

Training@1

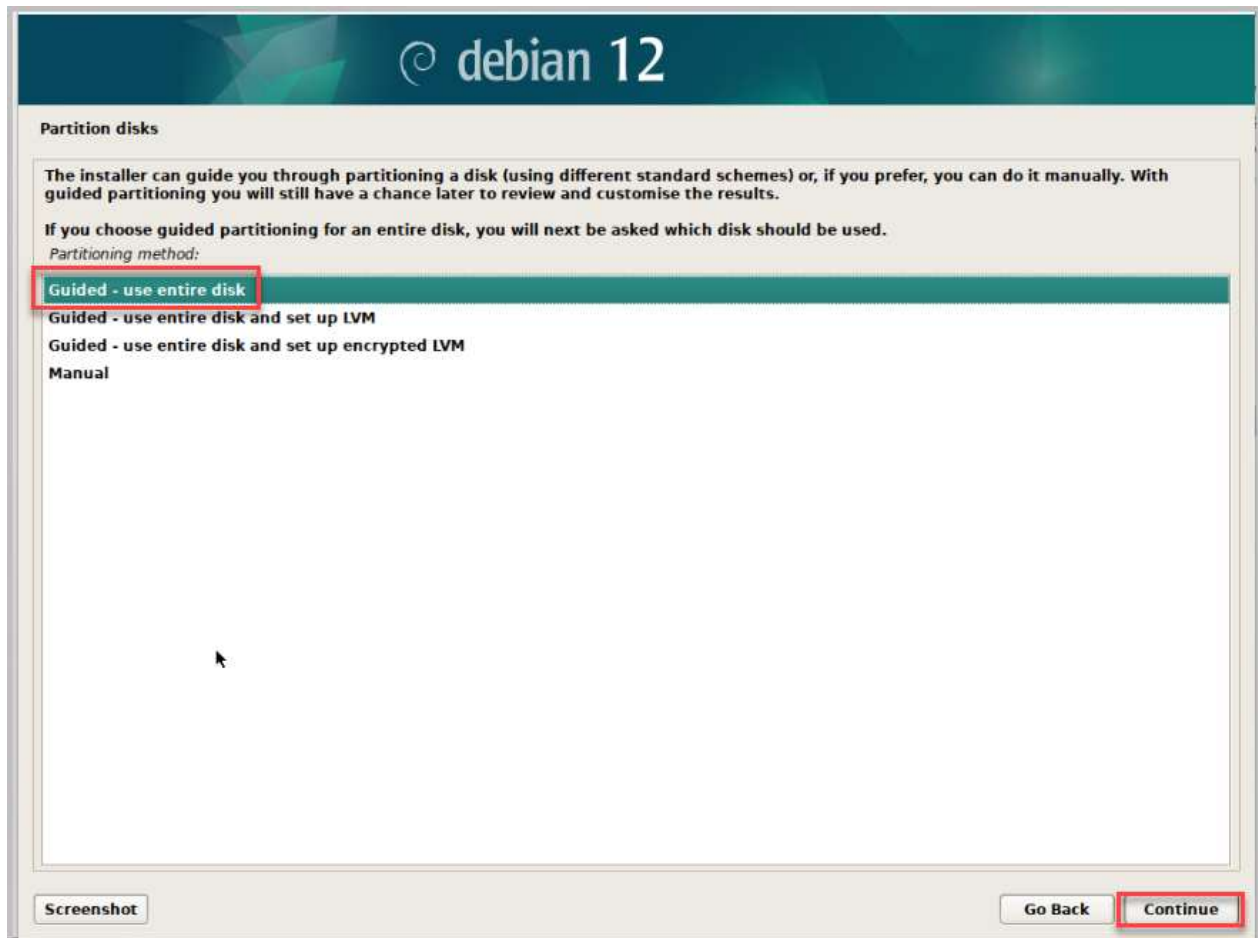
☒ Show Password in Clear

Screenshot Go Back Continue

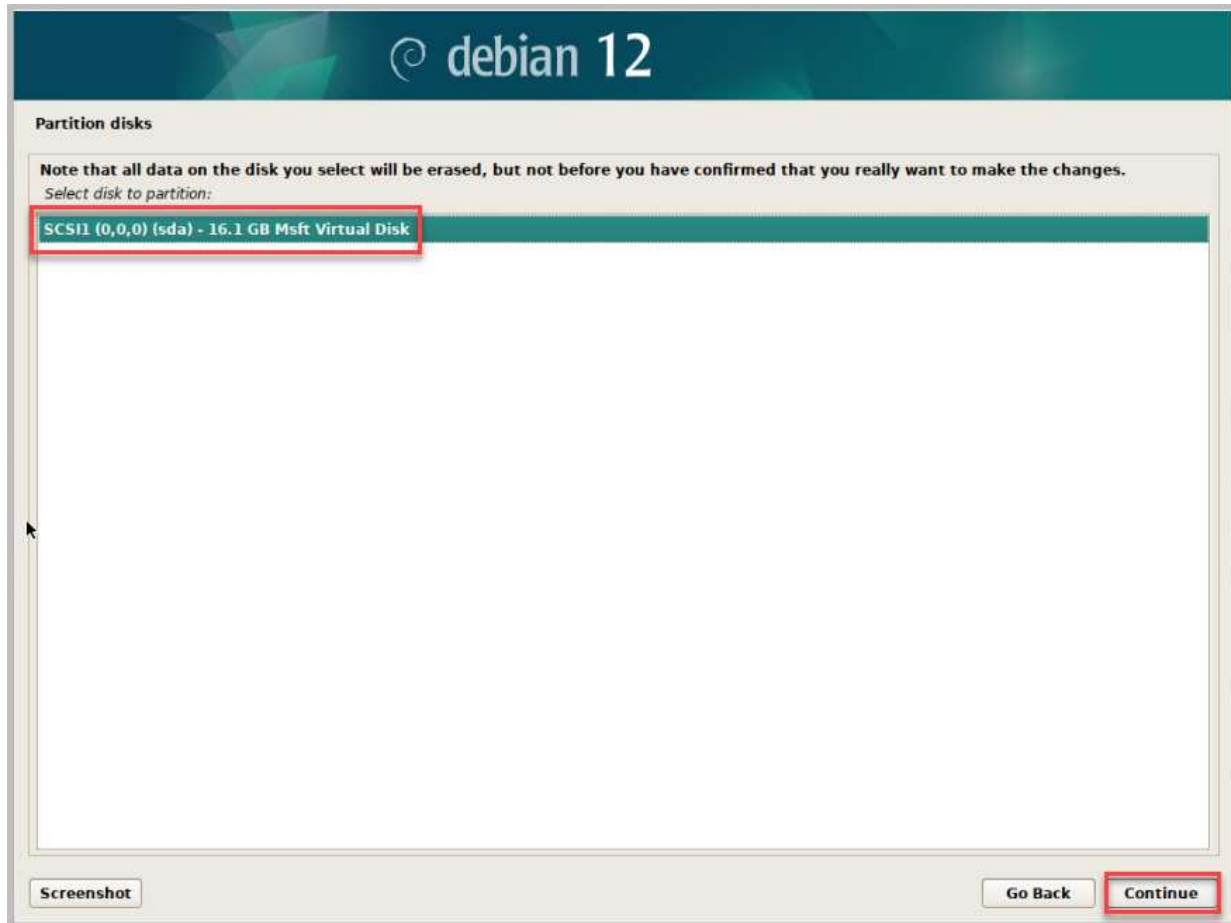
15. Leave the default selection of **Eastern** for the *desired time zone*. Select **Continue**.



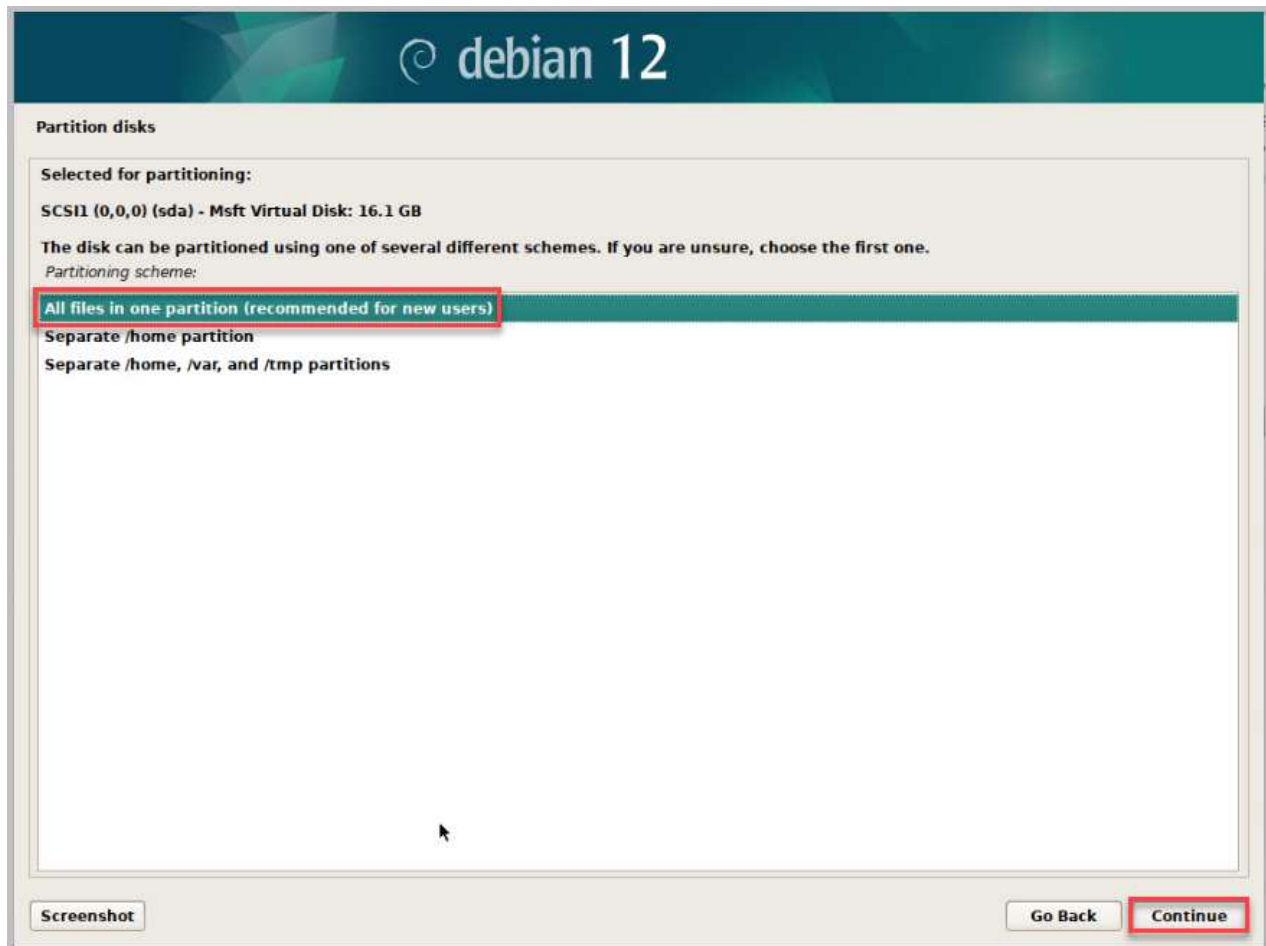
16. Leave the default selection of **Guided – use entire disk** for partitioning of the disk to use for installing the *Debian Operating System*. Select **Continue**.



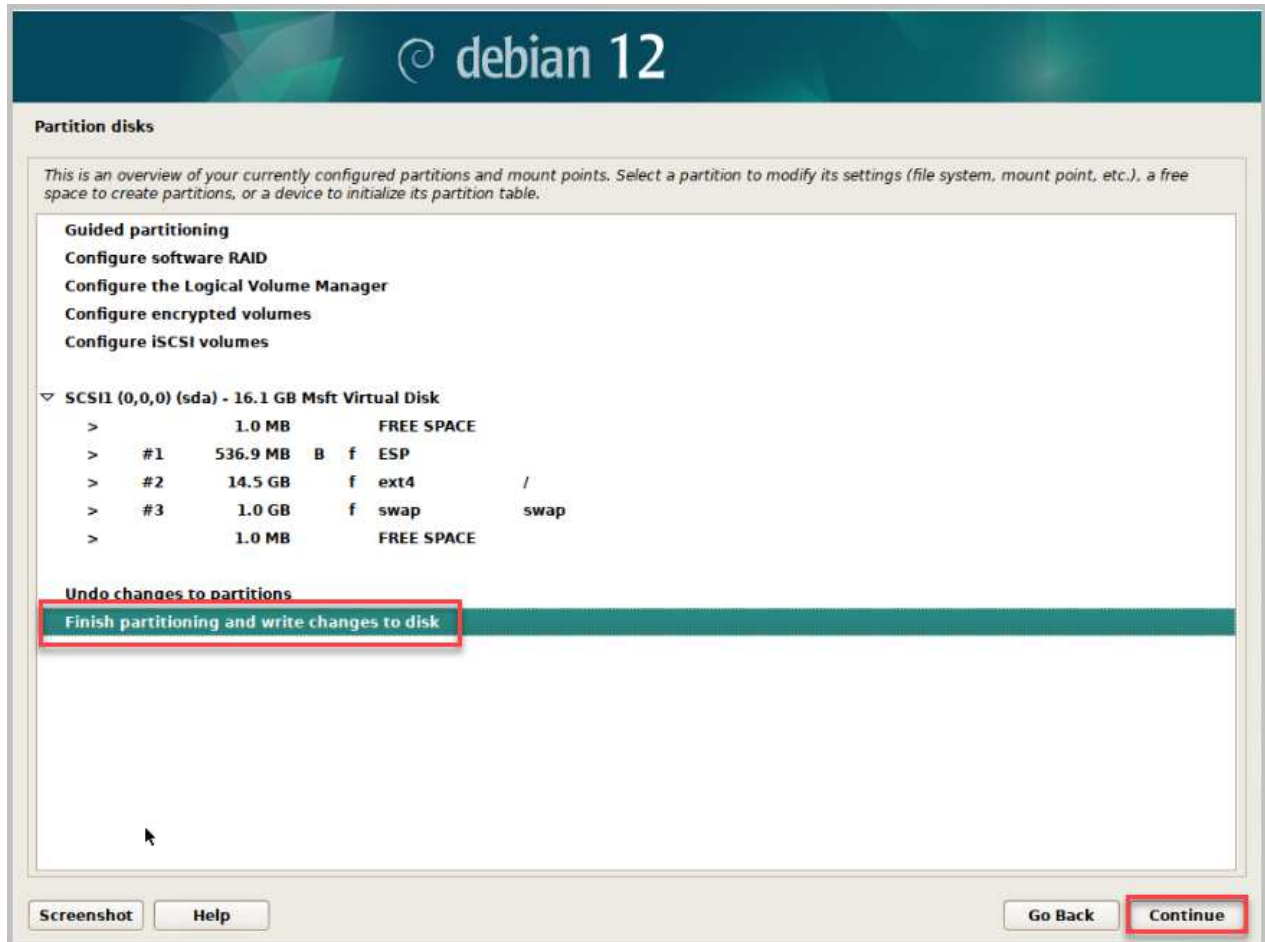
17. The disk that will be partitioned, **SCSI1 (0,0,0) (sda) – 16.1 GB Msft Virtual Disk** is already selected for you. Select **Continue**.



18. Leave the default selection of **All files in one partition (recommended for new users)**, as there is no need to split partitioning over multiple disks. Select **Continue**.



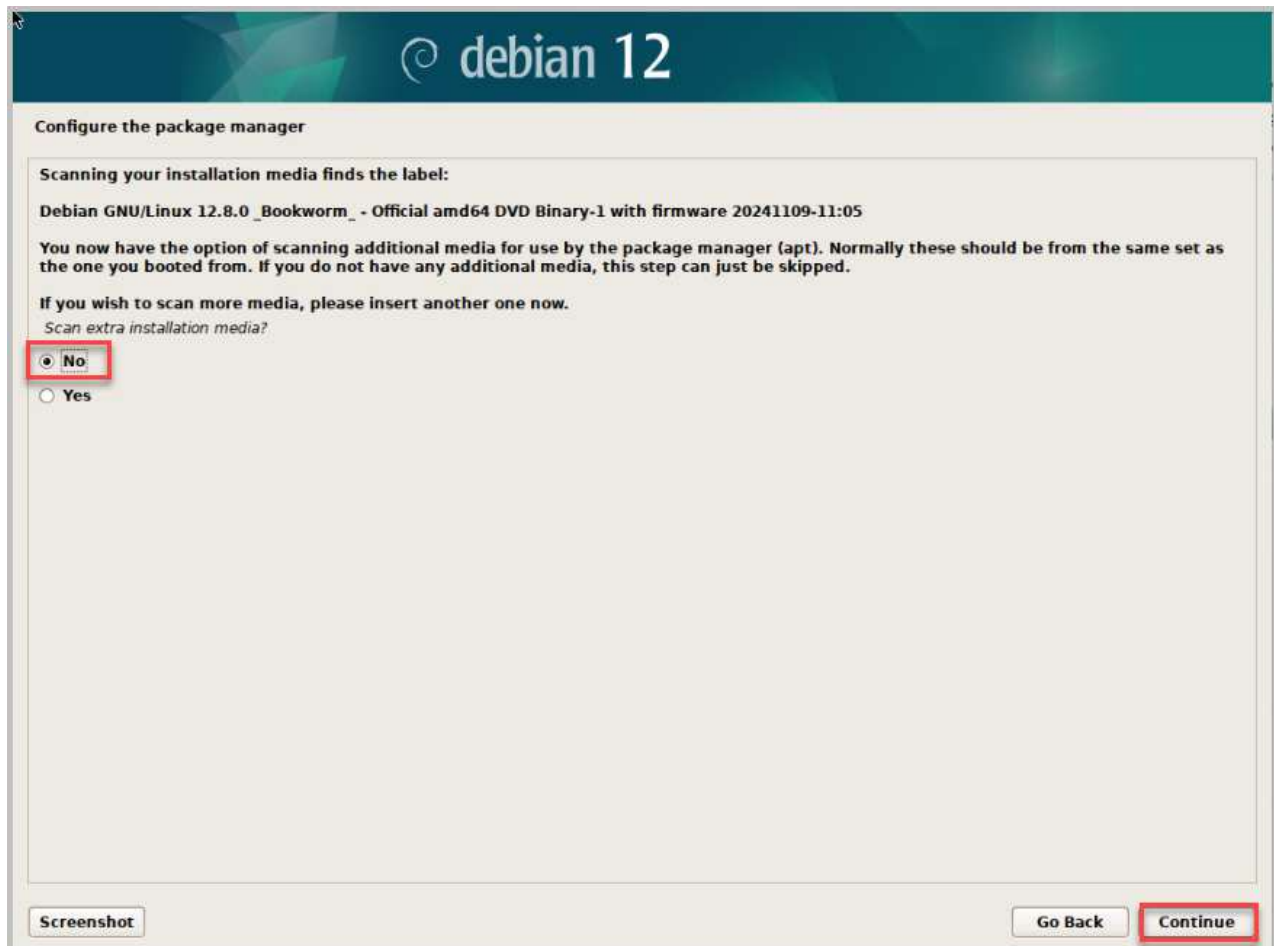
19. Read the overview of currently configured partitions and mount points. Leave the default selection of **Finish partitioning and write changes to disk**. Select **Continue**.



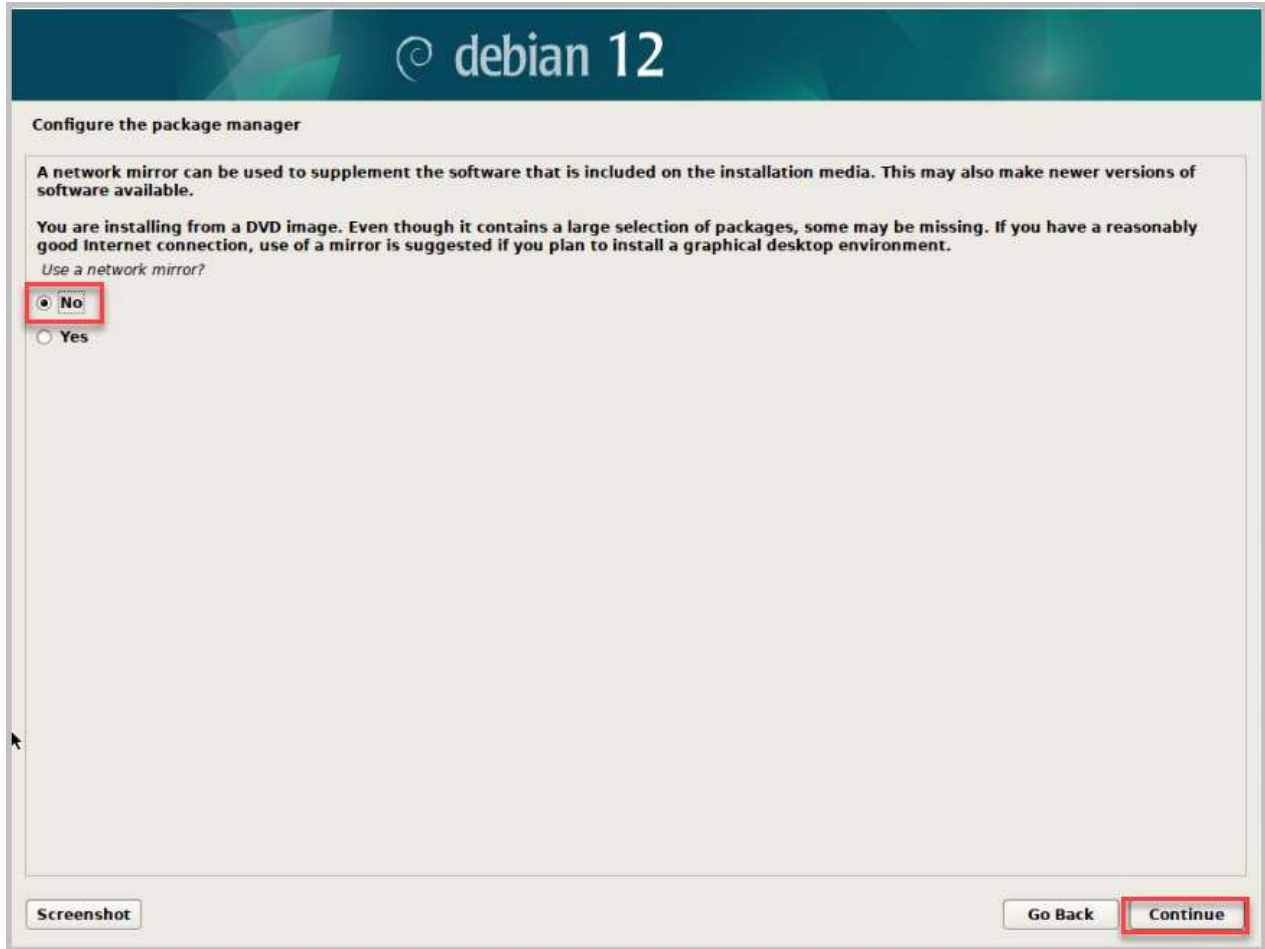
20. Select **Yes** to write the changes to disks, and select **Continue**.



21. Leave the default selection of **No** for configuring the package manager. There is no need to scan more media to complete this installation. Select **Continue**.



22. For the *Configure the package manager* page regarding a network mirror, leave the default selection of **No**. Select **Continue**.



The image shows the 'Configure the package manager' screen in the Debian 12 installer. The title bar at the top says 'debian 12'. The main heading is 'Configure the package manager'. Below it, there is a paragraph explaining that a network mirror can be used to supplement software on the installation media. Another paragraph states that since the user is installing from a DVD image, a network mirror is suggested if they plan to install a graphical desktop environment. The question 'Use a network mirror?' is followed by two radio button options: 'No' (which is selected and highlighted with a red box) and 'Yes'. At the bottom of the window, there are three buttons: 'Screenshot', 'Go Back', and 'Continue' (which is also highlighted with a red box).

© debian 12

Configure the package manager

A network mirror can be used to supplement the software that is included on the installation media. This may also make newer versions of software available.

You are installing from a DVD image. Even though it contains a large selection of packages, some may be missing. If you have a reasonably good Internet connection, use of a mirror is suggested if you plan to install a graphical desktop environment.

Use a network mirror?

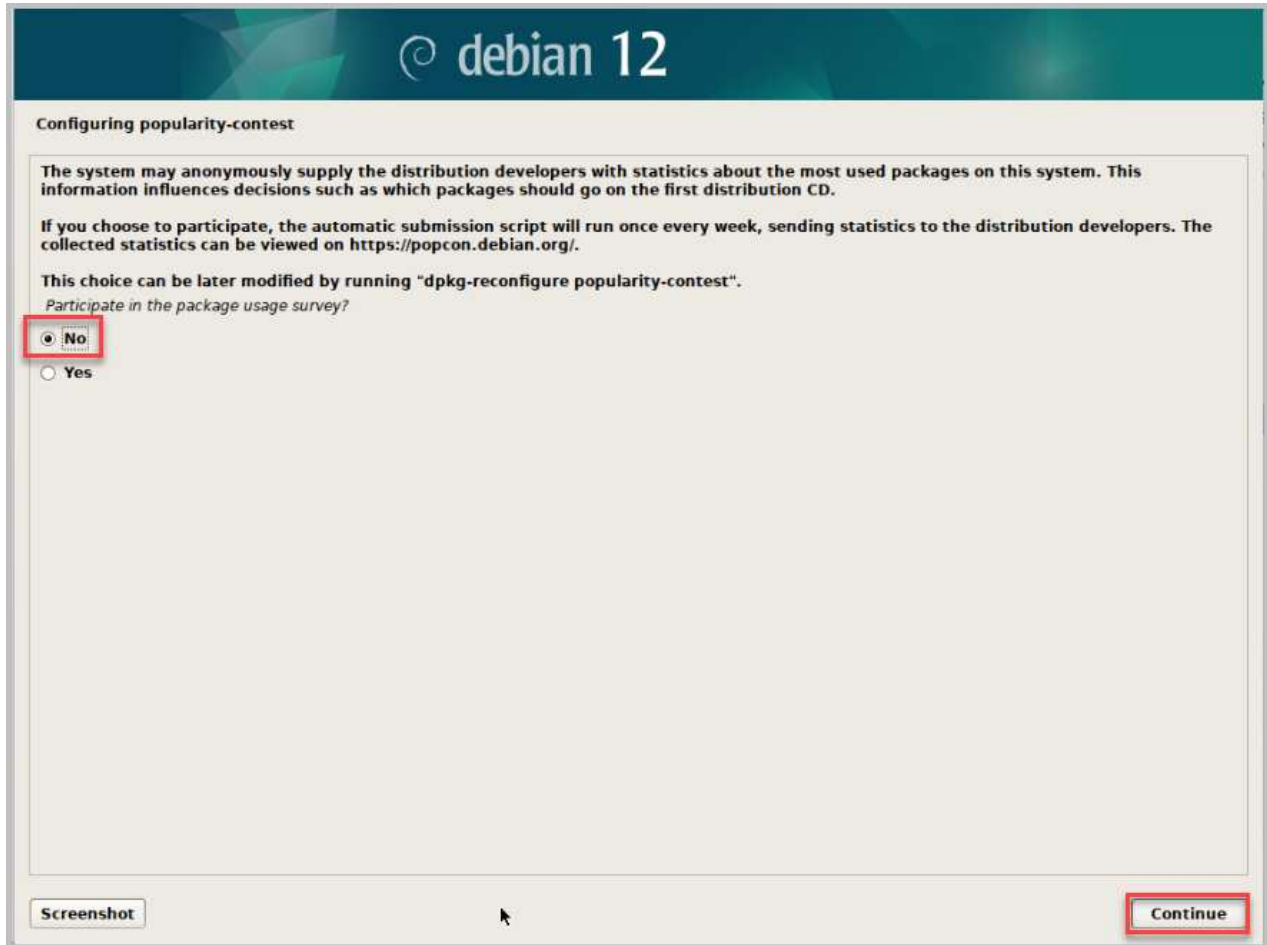
☒ **No**

☐ Yes

Screenshot

Go Back **Continue**

23. For the *Configuring popularity-contest*, leave the default selection of **No**. As the *New-DebianVM* does not have internet connectivity, there is no way to send any statistics about this VM to the Distribution Developers of the Debian OS. Select **Continue**.



© debian 12

Configuring popularity-contest

The system may anonymously supply the distribution developers with statistics about the most used packages on this system. This information influences decisions such as which packages should go on the first distribution CD.

If you choose to participate, the automatic submission script will run once every week, sending statistics to the distribution developers. The collected statistics can be viewed on <https://popcon.debian.org/>.

This choice can be later modified by running "dpkg-reconfigure popularity-contest".

Participate in the package usage survey?

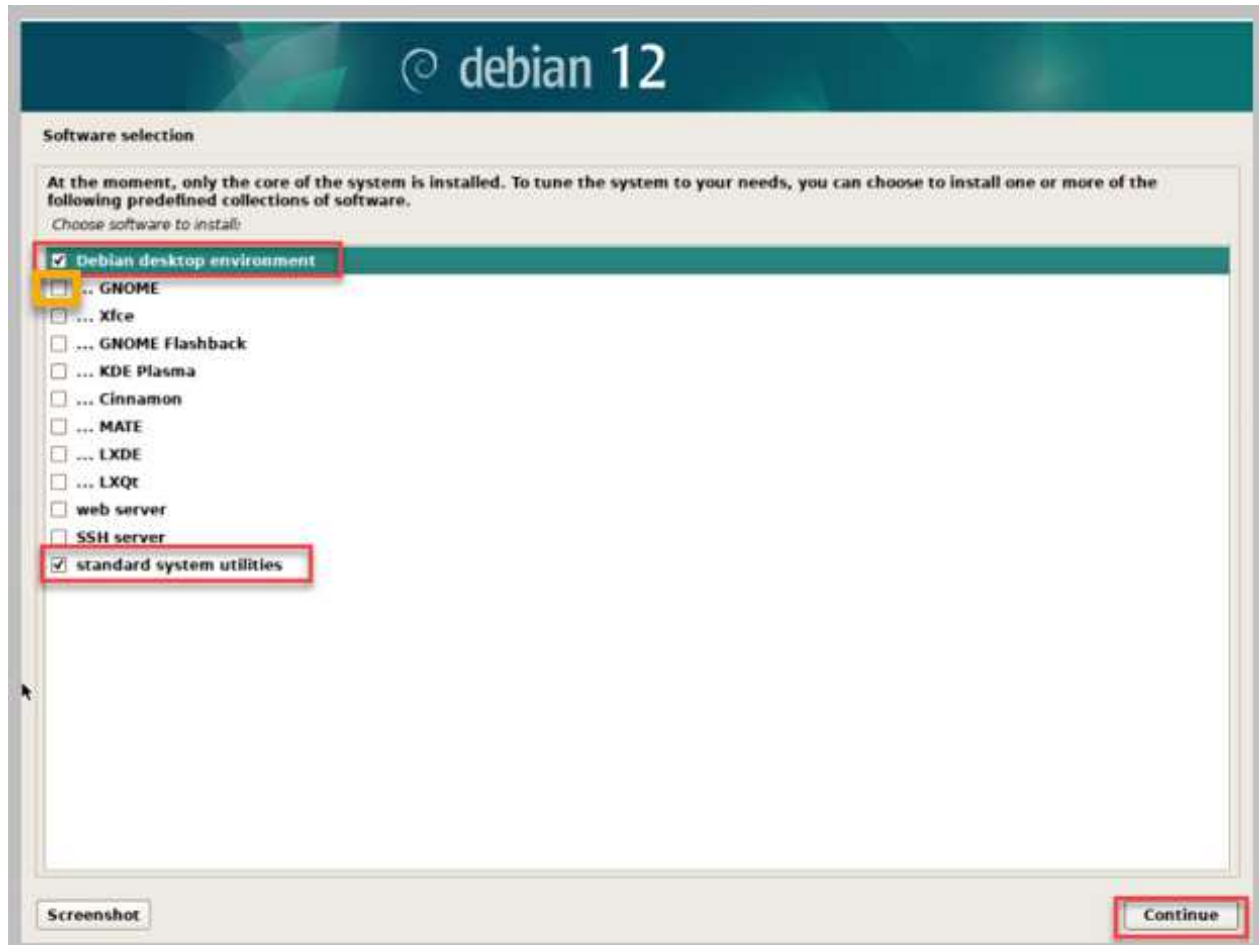
☒ No

☐ Yes

Screenshot

Continue

24. Leave the defaults for *Software selection*, **Debian desktop environment**, and **standard system utilities** as they are considered the core options needed to run successfully. Uncheck **GNOME** if it is checked. Select **Continue**.



**Core Options:****1. Debian Desktop Environment:**

- Installs a full-featured desktop environment like GNOME, KDE Plasma, Xfce, Cinnamon, MATE, LXDE, or LXQt.
- Provides a graphical user interface (GUI) for easy interaction.
- Suitable for general computing tasks.

2. Standard System Utilities:

- Install essential command-line tools for system administration and basic tasks.
- Ideal for server environments or users who prefer working in a text-based interface.

Desktop Environments:**1. GNOME:**

- Modern and feature-rich desktop environment.
- Known for its user-friendly interface and customization options.

2. Xfce:

- Lightweight and efficient desktop environment.
- Suitable for older hardware or users who prefer a minimalist approach.

3. GNOME Flashback:

- Traditional desktop environment based on GNOME 2.
- Familiar to users of older systems.

4. KDE Plasma:

- Highly customizable and feature-rich desktop environment.
- Offers a wide range of customization options and advanced features.

5. Cinnamon:

- Traditional desktop environment with a modern look and feel.
- Offers a balance between classic and modern design.

6. MATE:

- Fork of GNOME 2, providing a familiar and stable desktop environment.

7. LXDE:

- Extremely lightweight and resource-efficient desktop environment.
- Ideal for older hardware or netbooks.

8. LXQt:

- Modern and lightweight desktop environment.
- Combines elements of LXDE and Qt.

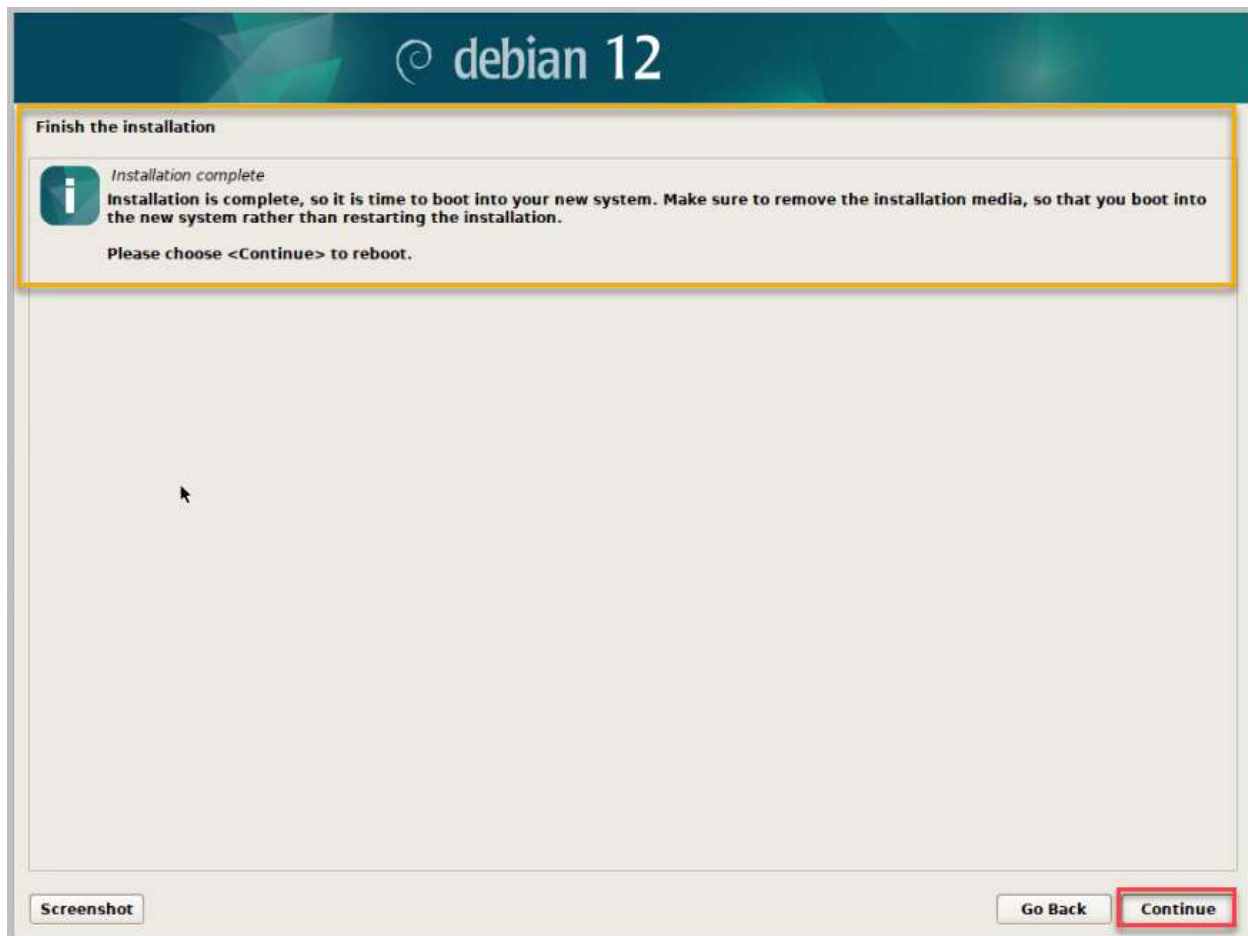
Additional Options:**1. Web Server:**

- Installs a web server like Apache or Nginx to host websites.

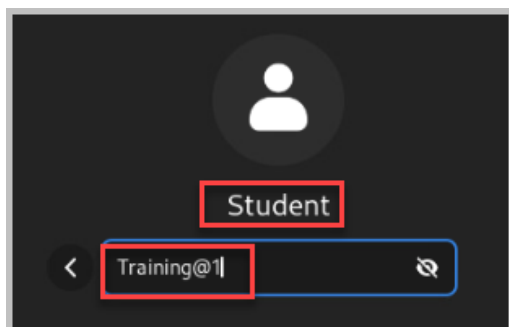
2. SSH Server:

- Enables secure remote access to your system via the SSH protocol.

25. This will take several minutes to complete. Once the *Installation is Complete*, select **Continue**.



26. The *New-DebianVM* will automatically reboot. Select **Student**. Enter password **Training@1** and hit **enter** on your keyboard to login to the *New-DebianVM*.



27. You have successfully installed and logged into the *New-DebianVM* virtual machine using *Hyper*

28. The lab is now complete; you may end the reservation.